

3.10 Inception networks

- **If residual networks was all about going deeper, the Inception networks is all about going wider**, as a way to increase the final network depth.
- The Inception network came to the world in 2014 when a group of researchers at Google published their paper "*Going Deeper with Convolutions*" (<https://arxiv.org/abs/1409.4842>).
- The main hallmark of this architecture is **building a deeper neural network while improving the utilization of the computing resources inside the network**.
- One particular incarnation of the Inception network is called *GoogLeNet*.

3.10.1. Inception modules

- There are some architectural decisions that you need to make for each layer when you are designing a CNN:
 - **The kernel size of the convolutional layer:**
 - When designing the convolutional layer, we find ourselves trying to pick and tune the kernel size of each layer that fits our dataset.
 - Smaller kernels capture finer details of the image, whereas bigger filters will leave out minute details.
 - **When to use the pooling layer:**
 - AlexNet uses pooling layers every one or two convolutional layers to downsize spatial features.
 - VGGNet applies pooling after every two, three, or four convolutional layers as the network gets deeper.
- Configuring the kernel size and positioning the pool layers are decisions we make mostly by trial and error and experiment with to get the optimal results.

why not let the model choose?

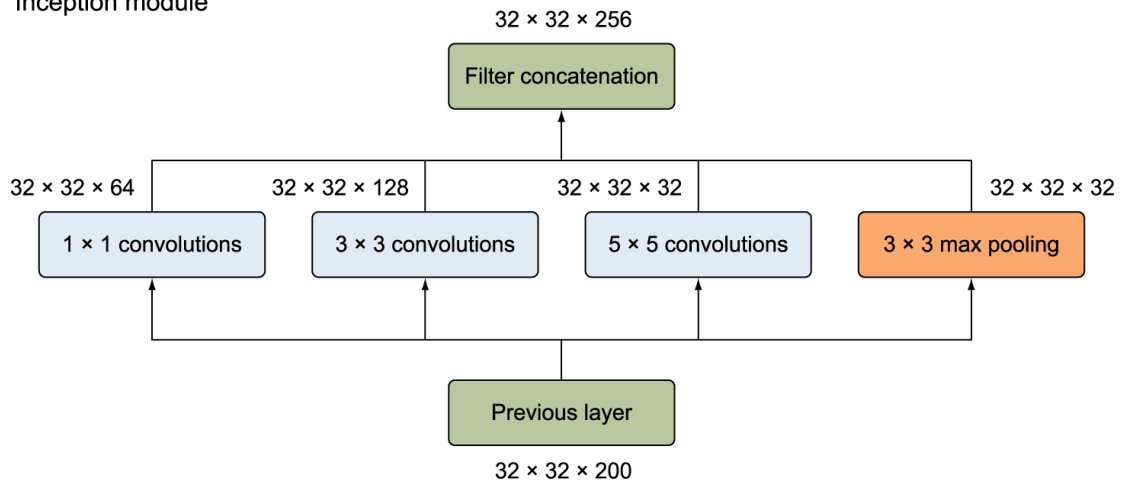
- **Instead of choosing a desired filter size in a convolutional layer and deciding where to place the pooling layers, let's apply all of them all together in one block and call it the inception module.**
- Let the next layer of the model gets to decide if (and how) to use each piece of information.
- The architecture is then developed by stacking the inception modules on top of each other.

Naïve inception module

- The inception module is a combination of four layers:
 - **1×1 convolutional layer**, looking for cross-channel correlations
 - **3×3 convolutional layer**, looking for finner spatial correlations
 - **5×5 convolutional layer**, looking for more-abstract spatial correlations

- **3×3 max-pooling layer**, trying to downsize spatial features.
- The outputs of these layers are concatenated into a single output volume forming the input of the next stage.
- We use the **same** padding so the output has the same length as the original input. We do that because we want the output to have width and height dimensions similar to the input to simplify the depth concatenation process.

Inception module



Problem with the naïve inception module

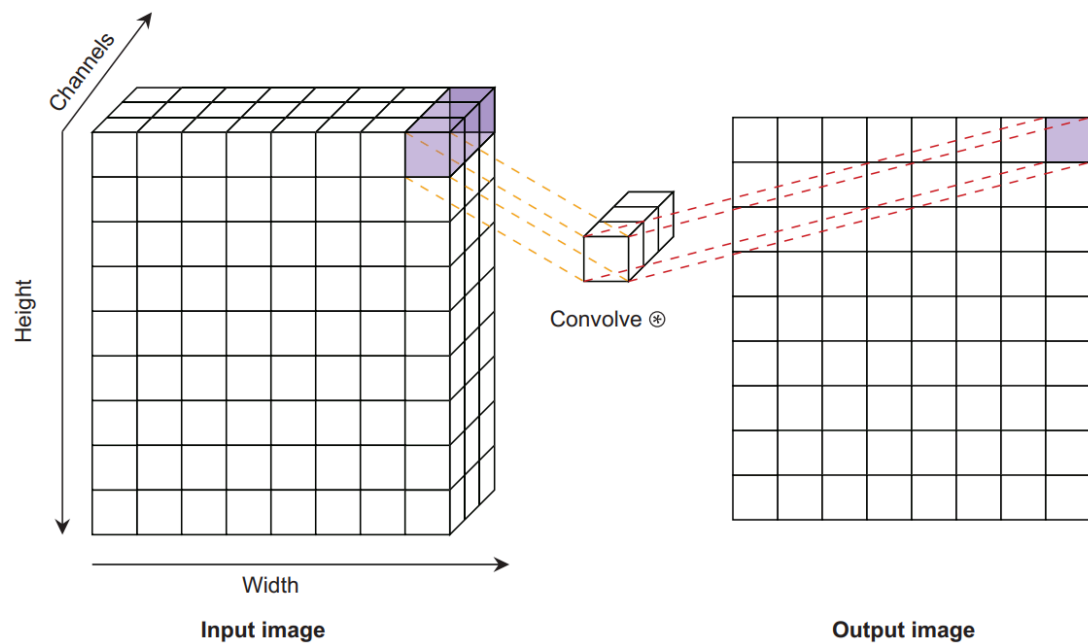
- We've drastically increased computational costs.
 - Large (e.g. 5×5) convolutional filters are inherently expensive to compute
 - Stacking multiple different filters side by side greatly increases the number of feature maps per layer.

3.10.2. Dimensionality reduction layer

1×1 Convolutions: Sharing and reshaping information in channels

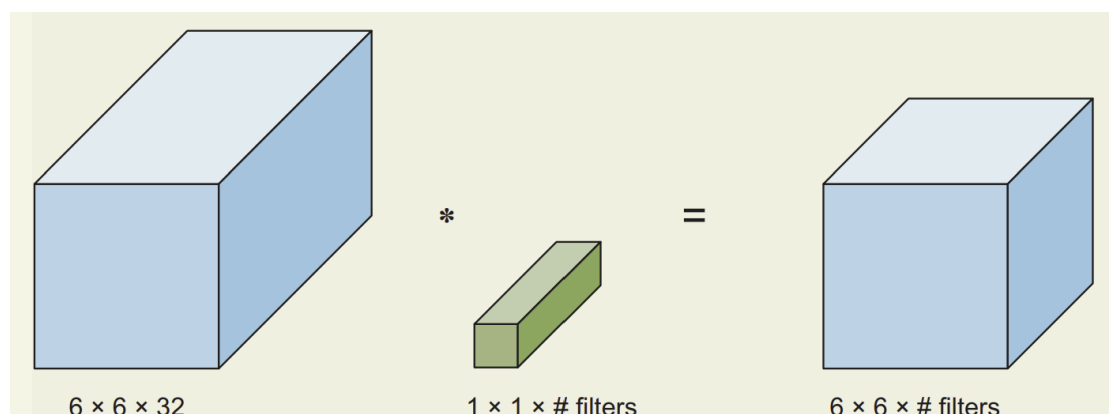
- Convolutions with a kernel size greater than one are used to capture *spatial* information.
- But a convolution with __a kernel size of 1 captures _no spatial information__.
- At first glance, this seems to make a convolution a useless operation.
- However, there can be valuable reasons to use a convolution with this kind of neighbor blindness:
 - One particular application is as a computationally **cheap operation to change the number of channels at a given layer**.
 - Another is that 1x1 convolutions **share information across channels**, rather than looking at neighboring locations.
- In summary, with a 1x1 convolution we are telling the network to focus on the patterns found at this location instead of having it try to build new spatial patterns.
- **Example:**
 - If we are processing an image, and one channel has learned to identify horizontal edges, another vertical edges, another edges at a 45-degree angle, and so on.

- If we want a channel to learn to identify any edge, we can accomplish that by looking just at the channel values (i.e., have any of these angle-dependent edge-detecting filters fired?) without considering the neighboring pixels.
- If such identifications are useful, 1×1 convolutions can help improve learning and reduce computational cost.



Use a 1×1 convolutional layer as a dimensionality reduction layer.

- As previously said, a 1×1 convolutional layer preserves the spatial dimensions (height and width) of the input volume but changes the number of channels of the volume (depth).
- We have already use them in the residual networks to adjust the depth of the residual connection.



Example of dimensionality reduction

- **Original**
 - Input volume with dimensions of $32 \times 32 \times 200$.
 - if we apply 32 filters with kernel size of 5×5 .

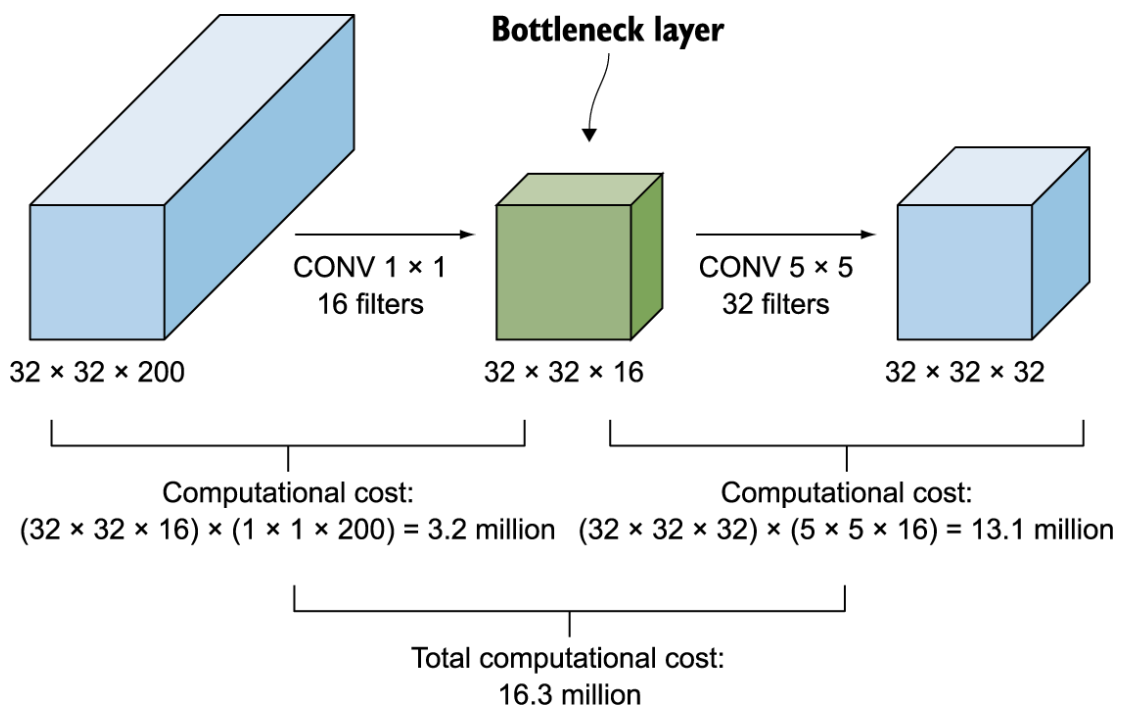
- Total number of multiplications: $32 \times 32 \times 200$ multiplied by $5 \times 5 \times 32 =$ more than 163 million operations.

- **With dimensionality reduction**

- We add a 1×1 convolutional layer with a depth of 16 to reduce the dimension volume from 200 to 16 channels.
- We can then apply the 5×5 convolutional layer on the output, which has much less depth.
- The total number of multiplications in this operation is 16.3 million, which is a tenth of the 163 million multiplications that we calculated without the reduce layers.

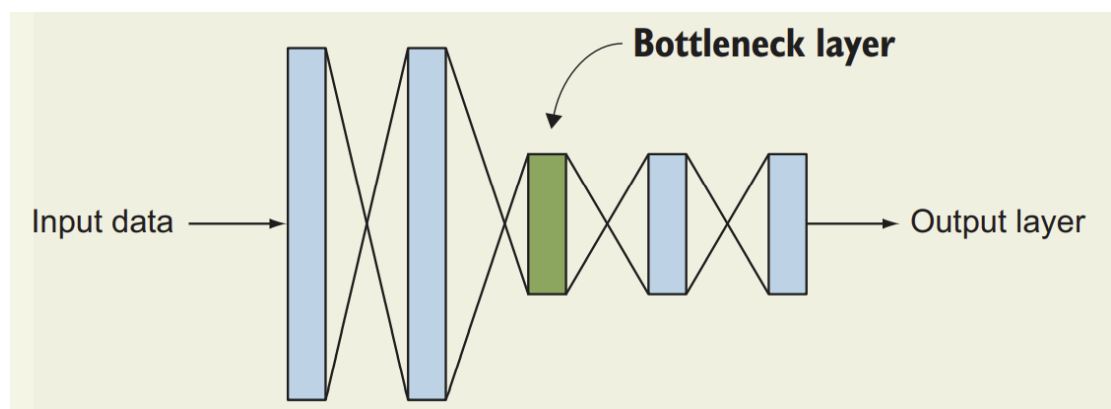
- **Conclusion**

- By dividing the convolution in two separate parts we obtain the same size at the output with much less operations.



Bottleneck layers

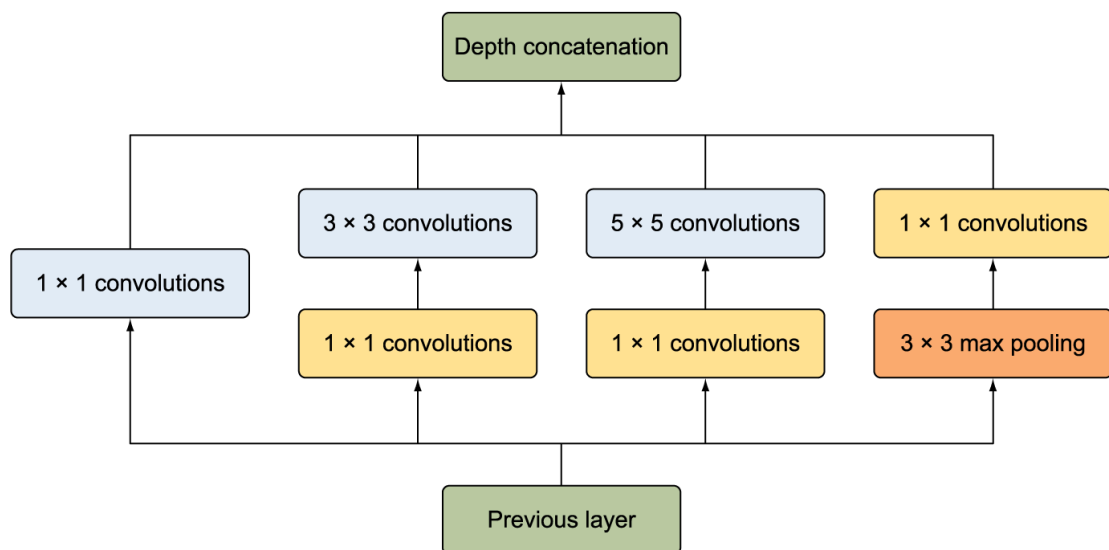
- The 1×1 convolutional layers are also known as bottleneck layers because by reducing the dimensionality of the network makes it look like a bottleneck:



3.10.3. Inception module with dimensionality reduction

- We maintain the 1×1 convolution to look for cross-channel correlations.
- We add a 1×1 convolutional layer before the bigger kernels (3×3 and 5×5), to reduce their depth, which in turn will reduce the number of operations.
- We will also add a 1×1 convolutional layer after the 3×3 max-pooling layer because pooling layers don't reduce the depth for their inputs. So, we will need to apply the reduce layer to their output before we do the concatenation.
- The design follows the practical intuition that visual information should be processed at various scales and then aggregated so that the next stage can abstract features from the different scales simultaneously.

Inception module with dimensionality reduction



3.10.4. Inception module in Keras

- Keras do not have pre-build inception modules but they are easy to create with the `Conv2D` and `MaxPool2D` layers.

```
In [1]: # Keras code for the inception_module

def inception_module(x, filters_1x1, filters_3x3_reduce, filters_3x3, filters_5x
    filters_5x5, filters_pool_proj, name=None):

    # 1x1 route
    conv_1x1 = Conv2D(filters_1x1, kernel_size=(1, 1), padding='same', activation='relu',
        kernel_initializer=kernel_init, bias_initializer=bias_init)(x)

    # 3x3 route = 1x1 CONV + 3x3 CONV
    pre_conv_3x3 = Conv2D(filters_3x3_reduce, kernel_size=(1, 1), padding='same',
        activation='relu', kernel_initializer=kernel_init,
        bias_initializer=bias_init)(x)
    conv_3x3 = Conv2D(filters_3x3, kernel_size=(3, 3), padding='same', activation='relu',
        kernel_initializer=kernel_init,
        bias_initializer=bias_init)(pre_conv_3x3)

    # 5x5 route = 1x1 CONV + 5x5 CONV
    pre_conv_5x5 = Conv2D(filters_5x5_reduce, kernel_size=(1, 1), padding='same',
        activation='relu', kernel_initializer=kernel_init,
        bias_initializer=bias_init)(x)
    conv_5x5 = Conv2D(filters_5x5, kernel_size=(5, 5), padding='same', activation='relu',
        kernel_initializer=kernel_init,
        bias_initializer=bias_init)(pre_conv_5x5)

    # Pooling route
    pool_proj = MaxPool2D(filters_pool_proj, kernel_size=(3, 3), padding='same',
        activation='relu', kernel_initializer=kernel_init,
        bias_initializer=bias_init)(x)

    # Depth concatenation
    return Concatenate(name=name)(conv_1x1, conv_3x3, conv_5x5, pool_proj)
```

```

# 5x5 route = 1x1 CONV + 5x5 CONV
pre_conv_5x5 = Conv2D(filters_5x5_reduce, kernel_size=(1, 1), padding='same',
                      activation='relu', kernel_initializer=kernel_init,
                      bias_initializer=bias_init)(x)

conv_5x5 = Conv2D(filters_5x5, kernel_size=(5, 5), padding='same', activation='relu',
                  kernel_initializer=kernel_init,
                  bias_initializer=bias_init)(pre_conv_5x5)

# pool route = POOL + 1x1 CONV
pool_proj = MaxPool2D((3, 3), strides=(1, 1), padding='same')(x)
pool_proj = Conv2D(filters_pool_proj, (1, 1), padding='same', activation='relu',
                  kernel_initializer=kernel_init, bias_initializer=bias_init)(pool_proj)

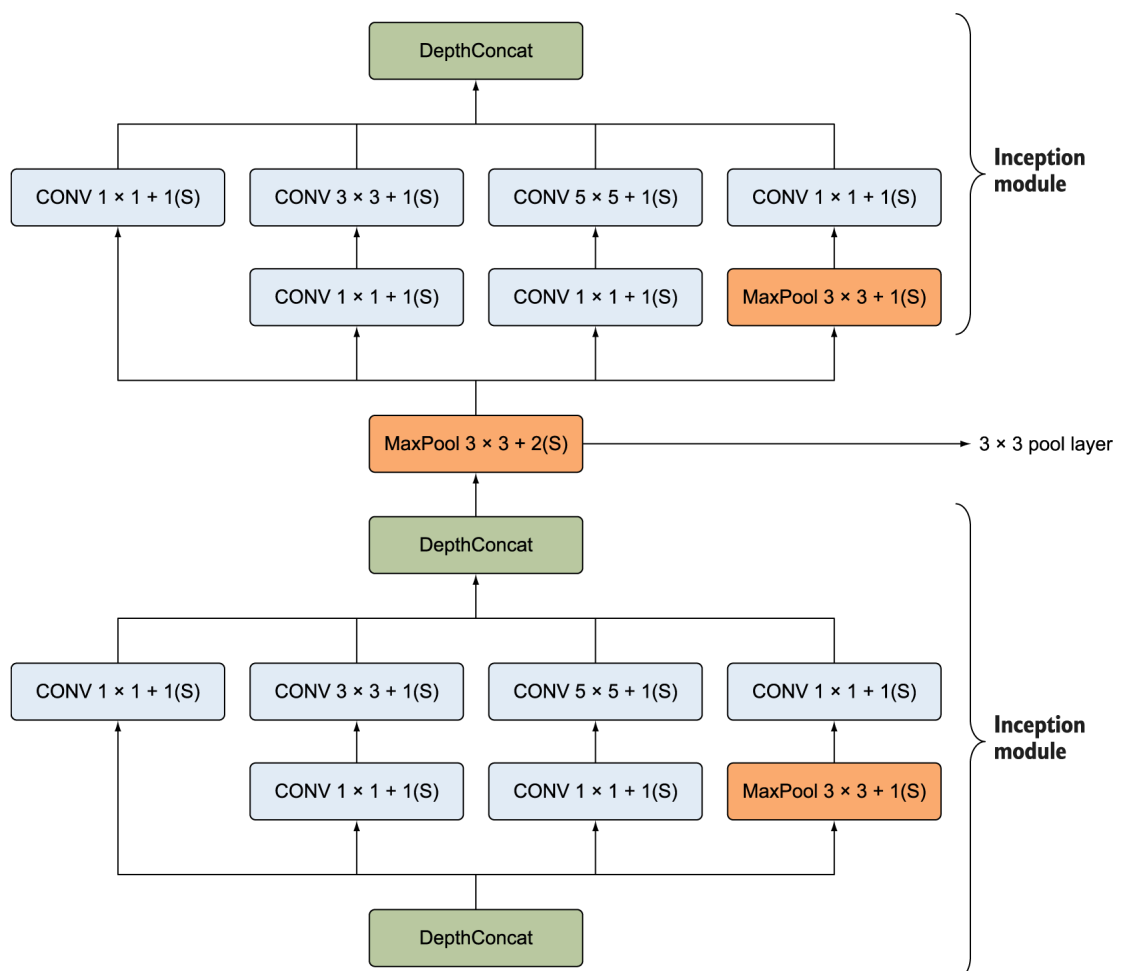
# final concatenation
output = concatenate([conv_1x1, conv_3x3, conv_5x5, pool_proj], axis=3, name='inception')

return output

```

3.10.5. Inception architecture

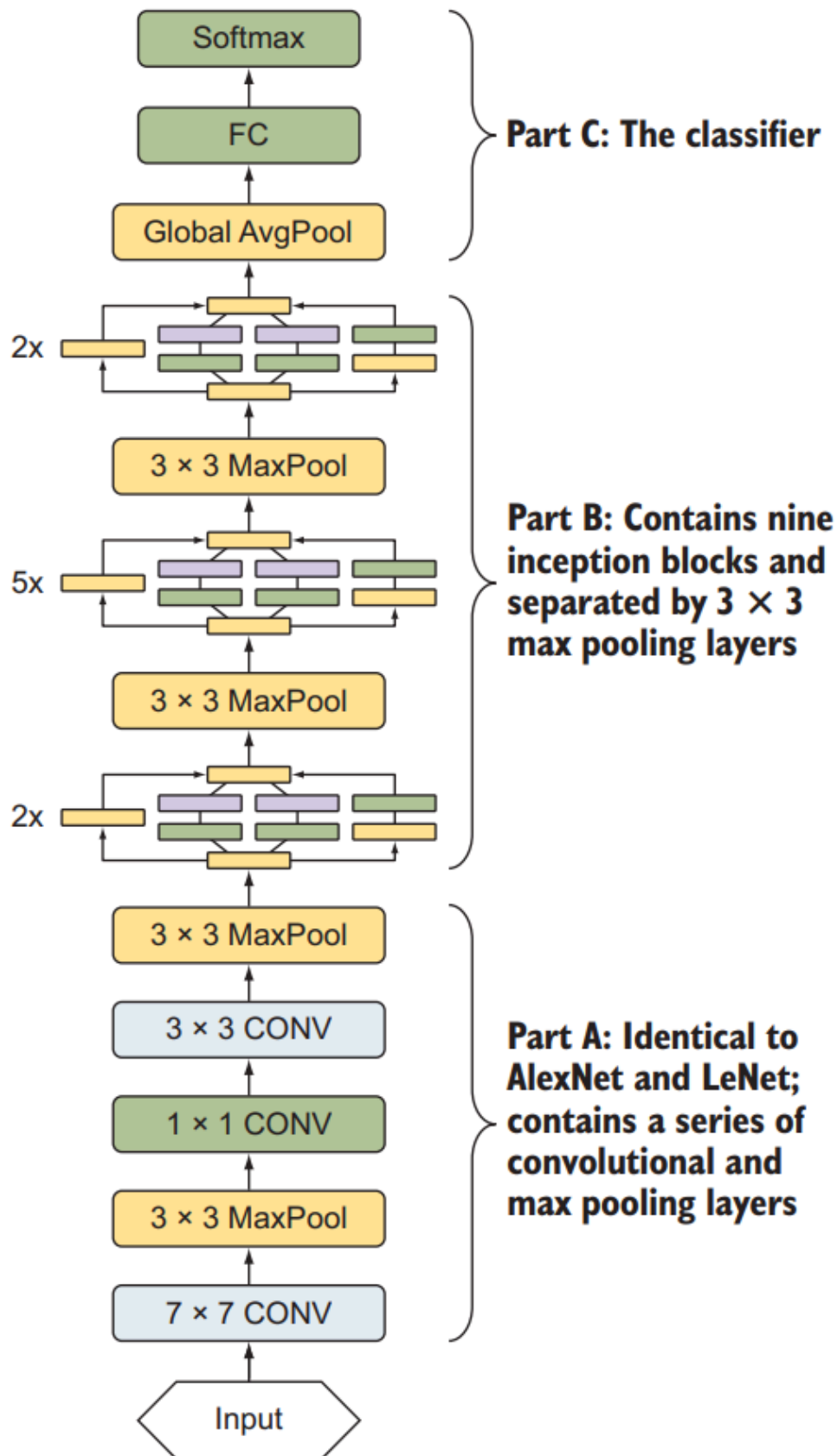
- The Inception network architecture consists of stacking inception modules on top of each other, and add a 3×3 pooling layer in between for downsampling.



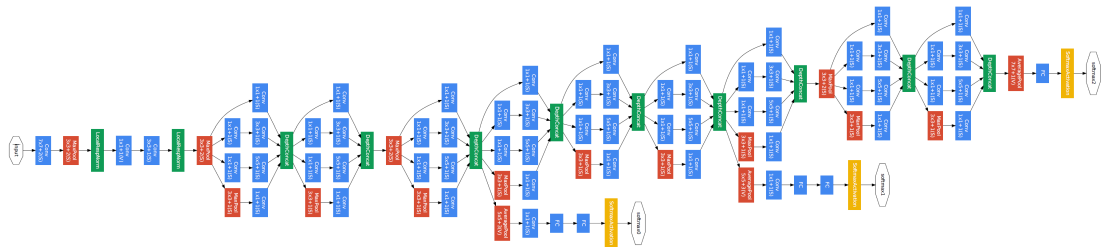
Inception V1 network (GoogLeNet)

- The full GoogLeNet model consists of three parts: the first part has the classical CNN architecture like AlexNet and LeNet, the second part is a stack of inception modules

and pooling layers, and the third part is the traditional fully connected classifiers.

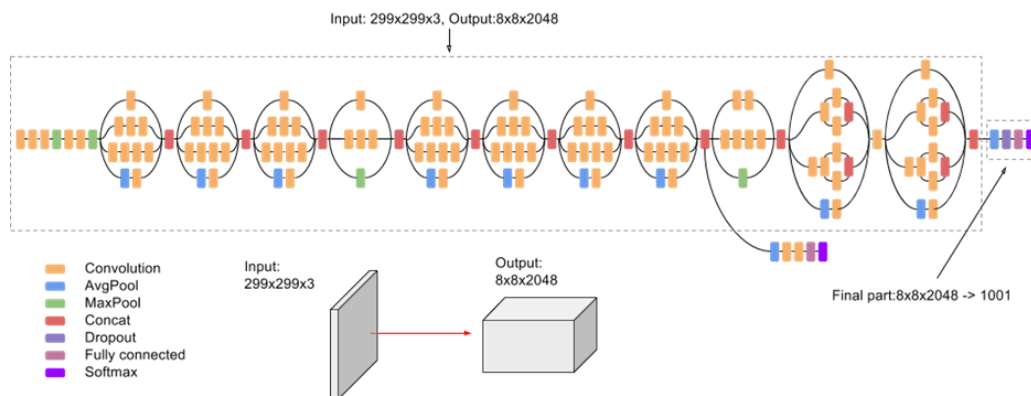


- Full image of Inception V1 network (GoogLeNet)



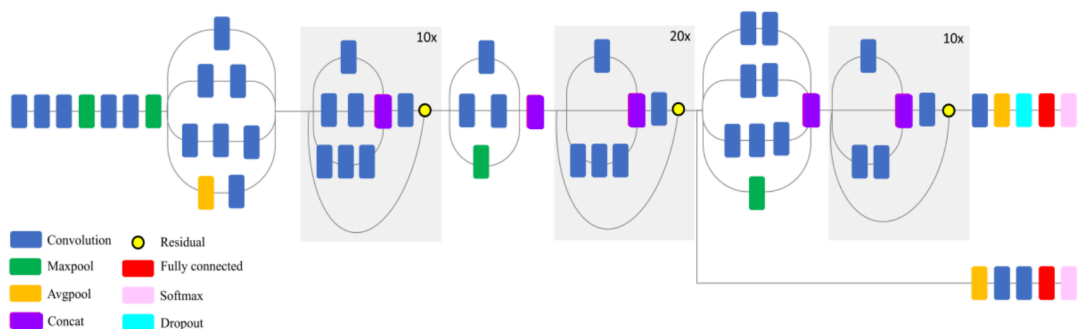
Inception V3 network

- The inception architecture received several updates.
- Version 3 is described here: <https://arxiv.org/abs/1512.00567>
- We can create pre-trained inception V3 models with the `inceptionv3` function. See: <https://keras.io/api/applications/inceptionv3/>



InceptionResNetV2

- Later versions included residual connections to the Inception architecture, such as the InceptionResNetV2 available in Keras: <https://keras.io/api/applications/inceptionresnetv2/>
- See: <https://arxiv.org/abs/1602.07261>



Cats and dogs example with a InceptionResNetV2 network with random initial weights

- Let's load the small cats-vs-dogs dataset again

```
In [2]: import keras
        from keras import layers
```

```
from keras.layers import Input, Conv2D, MaxPool2D, BatchNormalization, concaten
```

```
In [3]: import os, shutil, pathlib
#base_dir = pathlib.Path("/mnt/c/datasets/cats_vs_dogs_small")
base_dir = pathlib.Path("c:/datasets/dogs-vs-cats_small")

from keras.utils import image_dataset_from_directory
train_dataset = image_dataset_from_directory(
    base_dir / "train",
    image_size=(180, 180),
    batch_size=32)
validation_dataset = image_dataset_from_directory(
    base_dir / "validation",
    image_size=(180, 180),
    batch_size=32)
test_dataset = image_dataset_from_directory(
    base_dir / "test",
    image_size=(180, 180),
    batch_size=32)
```

Found 2000 files belonging to 2 classes.

Found 1000 files belonging to 2 classes.

Found 2000 files belonging to 2 classes.

```
In [4]: from keras.applications import InceptionResNetV2
```

```
In [5]: conv_base = InceptionResNetV2(
    weights=None,
    include_top=False)
```

WARNING:tensorflow:From C:\Users\eduar\anaconda3\envs\AP\Lib\site-packages\keras\src\backend\tensorflow\core.py:204: The name tf.placeholder is deprecated. Please use tf.compat.v1.placeholder instead.

```
In [6]: conv_base.summary()
```

Model: "inception_resnet_v2"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, None, None, 3)	0	-
conv2d (Conv2D)	(None, None, None, 32)	864	input_layer
batch_normalization (BatchNormalizatio...	(None, None, None, 32)	96	conv2d[0][0]
activation (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_1 (Conv2D)	(None, None, None, 32)	9,216	activation
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_1[0]
activation_1 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_2 (Conv2D)	(None, None, None, 64)	18,432	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_2[0]
activation_2 (Activation)	(None, None, None, 64)	0	batch_norma
max_pooling2d (MaxPooling2D)	(None, None, None, 64)	0	activation_
conv2d_3 (Conv2D)	(None, None, None, 80)	5,120	max_pooling
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 80)	240	conv2d_3[0]
activation_3 (Activation)	(None, None, None, 80)	0	batch_norma
conv2d_4 (Conv2D)	(None, None, None, 192)	138,240	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_4[0]
activation_4 (Activation)	(None, None, None, 192)	0	batch_norma
max_pooling2d_1 (MaxPooling2D)	(None, None, None, 192)	0	activation_
conv2d_8 (Conv2D)	(None, None, None, 64)	12,288	max_pooling
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_8[0]
activation_8	(None, None,	0	batch_norma

(Activation)	None, 64)		
conv2d_6 (Conv2D)	(None, None, None, 48)	9,216	max_pooling2d_6[0]
conv2d_9 (Conv2D)	(None, None, None, 96)	55,296	activation_9[0]
batch_normalization_6 (BatchNormalization)	(None, None, None, 48)	144	conv2d_6[0]
batch_normalization_9 (BatchNormalization)	(None, None, None, 96)	288	conv2d_9[0]
activation_6 (Activation)	(None, None, None, 48)	0	batch_normalization_6[0]
activation_9 (Activation)	(None, None, None, 96)	0	batch_normalization_9[0]
average_pooling2d_6 (AveragePooling2D)	(None, None, None, 192)	0	max_pooling2d_6[0]
conv2d_5 (Conv2D)	(None, None, None, 96)	18,432	max_pooling2d_5[0]
conv2d_7 (Conv2D)	(None, None, None, 64)	76,800	activation_7[0]
conv2d_10 (Conv2D)	(None, None, None, 96)	82,944	activation_10[0]
conv2d_11 (Conv2D)	(None, None, None, 64)	12,288	average_pooling2d_11[0]
batch_normalization_5 (BatchNormalization)	(None, None, None, 96)	288	conv2d_5[0]
batch_normalization_7 (BatchNormalization)	(None, None, None, 64)	192	conv2d_7[0]
batch_normalization_10 (BatchNormalization)	(None, None, None, 96)	288	conv2d_10[0]
batch_normalization_11 (BatchNormalization)	(None, None, None, 64)	192	conv2d_11[0]
activation_5 (Activation)	(None, None, None, 96)	0	batch_normalization_5[0]
activation_7 (Activation)	(None, None, None, 64)	0	batch_normalization_7[0]
activation_10 (Activation)	(None, None, None, 96)	0	batch_normalization_10[0]
activation_11 (Activation)	(None, None, None, 64)	0	batch_normalization_11[0]
mixed_5b (Concatenate)	(None, None, None, 320)	0	activation_5[0] activation_7[0] activation_10[0] activation_11[0]

conv2d_15 (Conv2D)	(None, None, None, 32)	10,240	mixed_5b[0]
batch_normalization_15 (BatchNormalization)	(None, None, None, 32)	96	conv2d_15[0]
activation_15 (Activation)	(None, None, None, 32)	0	batch_normalization_15[0]
conv2d_13 (Conv2D)	(None, None, None, 32)	10,240	mixed_5b[0]
conv2d_16 (Conv2D)	(None, None, None, 48)	13,824	activation_15[0]
batch_normalization_16 (BatchNormalization)	(None, None, None, 32)	96	conv2d_13[0]
batch_normalization_17 (BatchNormalization)	(None, None, None, 48)	144	conv2d_16[0]
activation_13 (Activation)	(None, None, None, 32)	0	batch_normalization_16[0]
activation_16 (Activation)	(None, None, None, 48)	0	batch_normalization_17[0]
conv2d_12 (Conv2D)	(None, None, None, 32)	10,240	mixed_5b[0]
conv2d_14 (Conv2D)	(None, None, None, 32)	9,216	activation_13[0]
conv2d_17 (Conv2D)	(None, None, None, 64)	27,648	activation_16[0]
batch_normalization_18 (BatchNormalization)	(None, None, None, 32)	96	conv2d_12[0]
batch_normalization_19 (BatchNormalization)	(None, None, None, 32)	96	conv2d_14[0]
batch_normalization_20 (BatchNormalization)	(None, None, None, 64)	192	conv2d_17[0]
activation_12 (Activation)	(None, None, None, 32)	0	batch_normalization_18[0]
activation_14 (Activation)	(None, None, None, 32)	0	batch_normalization_19[0]
activation_17 (Activation)	(None, None, None, 64)	0	batch_normalization_20[0]
block35_1_mixed (Concatenate)	(None, None, None, 128)	0	activation_12[0] activation_14[0] activation_17[0]
block35_1_conv (Conv2D)	(None, None, None, 320)	41,280	block35_1_mixed[0]
custom_scale_layer (CustomScaleLayer)	(None, None, None, 320)	0	mixed_5b[0] block35_1_conv[0]

block35_1_ac (Activation)	(None, None, None, 320)	0	custom_scal
conv2d_21 (Conv2D)	(None, None, None, 32)	10,240	block35_1_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_21[(
activation_21 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_19 (Conv2D)	(None, None, None, 32)	10,240	block35_1_a
conv2d_22 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_19[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_22[(
activation_19 (Activation)	(None, None, None, 32)	0	batch_norma
activation_22 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_18 (Conv2D)	(None, None, None, 32)	10,240	block35_1_a
conv2d_20 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_23 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_18[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_20[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_23[(
activation_18 (Activation)	(None, None, None, 32)	0	batch_norma
activation_20 (Activation)	(None, None, None, 32)	0	batch_norma
activation_23 (Activation)	(None, None, None, 64)	0	batch_norma
block35_2_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_2_conv (Conv2D)	(None, None, None, 320)	41,280	block35_2_r

custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_1_a block35_2_c
block35_2_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_27 (Conv2D)	(None, None, None, 32)	10,240	block35_2_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_27[(
activation_27 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_25 (Conv2D)	(None, None, None, 32)	10,240	block35_2_a
conv2d_28 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_25[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_28[(
activation_25 (Activation)	(None, None, None, 32)	0	batch_norma
activation_28 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_24 (Conv2D)	(None, None, None, 32)	10,240	block35_2_a
conv2d_26 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_29 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_24[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_26[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_29[(
activation_24 (Activation)	(None, None, None, 32)	0	batch_norma
activation_26 (Activation)	(None, None, None, 32)	0	batch_norma
activation_29 (Activation)	(None, None, None, 64)	0	batch_norma
block35_3_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_

block35_3_conv (Conv2D)	(None, None, None, 320)	41,280	block35_3_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_2_a block35_3_c
block35_3_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_33 (Conv2D)	(None, None, None, 32)	10,240	block35_3_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_33[(
activation_33 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_31 (Conv2D)	(None, None, None, 32)	10,240	block35_3_a
conv2d_34 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_31[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_34[(
activation_31 (Activation)	(None, None, None, 32)	0	batch_norma
activation_34 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_30 (Conv2D)	(None, None, None, 32)	10,240	block35_3_a
conv2d_32 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_35 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_30[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_32[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_35[(
activation_30 (Activation)	(None, None, None, 32)	0	batch_norma
activation_32 (Activation)	(None, None, None, 32)	0	batch_norma
activation_35 (Activation)	(None, None, None, 64)	0	batch_norma
block35_4_mixed	(None, None,	0	activation_

(Concatenate)	(None, 128)		activation_ activation_
block35_4_conv (Conv2D)	(None, None, None, 320)	41,280	block35_4_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_3_a block35_4_c
block35_4_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_39 (Conv2D)	(None, None, None, 32)	10,240	block35_4_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_39[(
activation_39 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_37 (Conv2D)	(None, None, None, 32)	10,240	block35_4_a
conv2d_40 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_37[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_40[(
activation_37 (Activation)	(None, None, None, 32)	0	batch_norma
activation_40 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_36 (Conv2D)	(None, None, None, 32)	10,240	block35_4_a
conv2d_38 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_41 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_36[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_38[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_41[(
activation_36 (Activation)	(None, None, None, 32)	0	batch_norma
activation_38 (Activation)	(None, None, None, 32)	0	batch_norma
activation_41	(None, None,	0	batch_norma

(Activation)	(None, 64)		
block35_5_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_5_conv (Conv2D)	(None, None, None, 320)	41,280	block35_5_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_4_a block35_5_c
block35_5_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_45 (Conv2D)	(None, None, None, 32)	10,240	block35_5_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_45[(
activation_45 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_43 (Conv2D)	(None, None, None, 32)	10,240	block35_5_a
conv2d_46 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_43[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_46[(
activation_43 (Activation)	(None, None, None, 32)	0	batch_norma
activation_46 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_42 (Conv2D)	(None, None, None, 32)	10,240	block35_5_a
conv2d_44 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_47 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_42[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_44[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_47[(
activation_42 (Activation)	(None, None, None, 32)	0	batch_norma
activation_44	(None, None,	0	batch_norma

(Activation)	(None, 32)		
activation_47 (Activation)	(None, None, None, 64)	0	batch_norma
block35_6_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_6_conv (Conv2D)	(None, None, None, 320)	41,280	block35_6_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_5_a block35_6_c
block35_6_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_51 (Conv2D)	(None, None, None, 32)	10,240	block35_6_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_51[(
activation_51 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_49 (Conv2D)	(None, None, None, 32)	10,240	block35_6_a
conv2d_52 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_49[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_52[(
activation_49 (Activation)	(None, None, None, 32)	0	batch_norma
activation_52 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_48 (Conv2D)	(None, None, None, 32)	10,240	block35_6_a
conv2d_50 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_53 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_48[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_50[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_53[(
activation_48	(None, None,	0	batch_norma

(Activation)	(None, 32)		
activation_50 (Activation)	(None, None, None, 32)	0	batch_norma
activation_53 (Activation)	(None, None, None, 64)	0	batch_norma
block35_7_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_7_conv (Conv2D)	(None, None, None, 320)	41,280	block35_7_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_6_a block35_7_c
block35_7_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_57 (Conv2D)	(None, None, None, 32)	10,240	block35_7_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_57[(
activation_57 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_55 (Conv2D)	(None, None, None, 32)	10,240	block35_7_a
conv2d_58 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_55[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_58[(
activation_55 (Activation)	(None, None, None, 32)	0	batch_norma
activation_58 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_54 (Conv2D)	(None, None, None, 32)	10,240	block35_7_a
conv2d_56 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_59 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_54[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_56[(
batch_normalizatio...	(None, None,	192	conv2d_59[(

(BatchNormalizatio...	None, 64)		
activation_54 (Activation)	(None, None, None, 32)	0	batch_norma
activation_56 (Activation)	(None, None, None, 32)	0	batch_norma
activation_59 (Activation)	(None, None, None, 64)	0	batch_norma
block35_8_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_8_conv (Conv2D)	(None, None, None, 320)	41,280	block35_8_r
custom_scale_layer...	(None, None, None, 320)	0	block35_7_a block35_8_c
block35_8_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_63 (Conv2D)	(None, None, None, 32)	10,240	block35_8_a
batch_normalizatio...	(None, None, None, 32)	96	conv2d_63[(
activation_63 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_61 (Conv2D)	(None, None, None, 32)	10,240	block35_8_a
conv2d_64 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio...	(None, None, None, 32)	96	conv2d_61[(
batch_normalizatio...	(None, None, None, 48)	144	conv2d_64[(
activation_61 (Activation)	(None, None, None, 32)	0	batch_norma
activation_64 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_60 (Conv2D)	(None, None, None, 32)	10,240	block35_8_a
conv2d_62 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_65 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio...	(None, None, None, 32)	96	conv2d_60[(
batch_normalizatio...	(None, None,	96	conv2d_62[(

(BatchNormalizatio...	(None, 32)		
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_65[(
activation_60 (Activation)	(None, None, None, 32)	0	batch_norma
activation_62 (Activation)	(None, None, None, 32)	0	batch_norma
activation_65 (Activation)	(None, None, None, 64)	0	batch_norma
block35_9_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_9_conv (Conv2D)	(None, None, None, 320)	41,280	block35_9_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_8_a block35_9_c
block35_9_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_69 (Conv2D)	(None, None, None, 32)	10,240	block35_9_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_69[(
activation_69 (Activation)	(None, None, None, 32)	0	batch_norma
conv2d_67 (Conv2D)	(None, None, None, 32)	10,240	block35_9_a
conv2d_70 (Conv2D)	(None, None, None, 48)	13,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_67[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 48)	144	conv2d_70[(
activation_67 (Activation)	(None, None, None, 32)	0	batch_norma
activation_70 (Activation)	(None, None, None, 48)	0	batch_norma
conv2d_66 (Conv2D)	(None, None, None, 32)	10,240	block35_9_a
conv2d_68 (Conv2D)	(None, None, None, 32)	9,216	activation_
conv2d_71 (Conv2D)	(None, None, None, 64)	27,648	activation_
batch_normalizatio...	(None, None,	96	conv2d_66[(

(BatchNormalizatio...	(None, 32)		
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 32)	96	conv2d_68[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 64)	192	conv2d_71[(
activation_66 (Activation)	(None, None, None, 32)	0	batch_norma
activation_68 (Activation)	(None, None, None, 32)	0	batch_norma
activation_71 (Activation)	(None, None, None, 64)	0	batch_norma
block35_10_mixed (Concatenate)	(None, None, None, 128)	0	activation_ activation_ activation_
block35_10_conv (Conv2D)	(None, None, None, 320)	41,280	block35_10_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 320)	0	block35_9_& block35_10_
block35_10_ac (Activation)	(None, None, None, 320)	0	custom_sca1
conv2d_73 (Conv2D)	(None, None, None, 256)	81,920	block35_10_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_73[(
activation_73 (Activation)	(None, None, None, 256)	0	batch_norma
conv2d_74 (Conv2D)	(None, None, None, 256)	589,824	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_74[(
activation_74 (Activation)	(None, None, None, 256)	0	batch_norma
conv2d_72 (Conv2D)	(None, None, None, 384)	1,105,920	block35_10_
conv2d_75 (Conv2D)	(None, None, None, 384)	884,736	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 384)	1,152	conv2d_72[(
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 384)	1,152	conv2d_75[(
activation_72 (Activation)	(None, None, None, 384)	0	batch_norma
activation_75	(None, None,	0	batch_norma

(Activation)	(None, 384)		
max_pooling2d_2 (MaxPooling2D)	(None, None, None, 320)	0	block35_10_
mixed_6a (Concatenate)	(None, None, None, 1088)	0	activation_ activation_ max_pooling
conv2d_77 (Conv2D)	(None, None, None, 128)	139,264	mixed_6a[0]
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_77[0]
activation_77 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_78 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_78[0]
activation_78 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_76 (Conv2D)	(None, None, None, 192)	208,896	mixed_6a[0]
conv2d_79 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_76[0]
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_79[0]
activation_76 (Activation)	(None, None, None, 192)	0	batch_norma
activation_79 (Activation)	(None, None, None, 192)	0	batch_norma
block17_1_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_1_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_1_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	mixed_6a[0] block17_1_c
block17_1_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_81 (Conv2D)	(None, None, None, 128)	139,264	block17_1_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_81[0]
activation_81	(None, None,	0	batch_norma

(Activation)	None, 128)		
conv2d_82 (Conv2D)	(None, None, None, 160)	143,360	activation_82 (Activation)
batch_normalization_82 (BatchNormalization)	(None, None, None, 160)	480	conv2d_82[0] (Conv2D)
activation_82 (Activation)	(None, None, None, 160)	0	batch_normalization_82 (BatchNormalization)
conv2d_80 (Conv2D)	(None, None, None, 192)	208,896	block17_1_activation_80 (Activation)
conv2d_83 (Conv2D)	(None, None, None, 192)	215,040	activation_82[0] (Conv2D)
batch_normalization_83 (BatchNormalization)	(None, None, None, 192)	576	conv2d_80[0] (Conv2D)
batch_normalization_80 (BatchNormalization)	(None, None, None, 192)	576	conv2d_83[0] (Conv2D)
activation_80 (Activation)	(None, None, None, 192)	0	batch_normalization_80 (BatchNormalization)
activation_83 (Activation)	(None, None, None, 192)	0	batch_normalization_83 (BatchNormalization)
block17_2_mixed (Concatenate)	(None, None, None, 384)	0	activation_80[0] (Conv2D)
block17_2_conv (Conv2D)	(None, None, None, 1088)	418,880	activation_83[0] (Conv2D)
custom_scale_layer_1 (CustomScaleLayer)	(None, None, None, 1088)	0	block17_2_mixed (Concatenate)
block17_2_activation_83 (Activation)	(None, None, None, 1088)	0	block17_2_conv[0] (Conv2D)
conv2d_85 (Conv2D)	(None, None, None, 128)	139,264	custom_scale_layer_1 (CustomScaleLayer)
batch_normalization_85 (BatchNormalization)	(None, None, None, 128)	384	block17_2_activation_83 (Activation)
activation_85 (Activation)	(None, None, None, 128)	0	conv2d_85[0] (Conv2D)
conv2d_86 (Conv2D)	(None, None, None, 160)	143,360	activation_85 (Activation)
batch_normalization_86 (BatchNormalization)	(None, None, None, 160)	480	conv2d_86[0] (Conv2D)
activation_86 (Activation)	(None, None, None, 160)	0	batch_normalization_86 (BatchNormalization)
conv2d_84 (Conv2D)	(None, None, None, 192)	208,896	activation_86 (Activation)
conv2d_87 (Conv2D)	(None, None, None, 192)	215,040	conv2d_84[0] (Conv2D)

batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_84[0]
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_87[0]
activation_84 (Activation)	(None, None, None, 192)	0	batch_norma
activation_87 (Activation)	(None, None, None, 192)	0	batch_norma
block17_3_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_3_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_3_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_2_a block17_3_c
block17_3_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_89 (Conv2D)	(None, None, None, 128)	139,264	block17_3_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_89[0]
activation_89 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_90 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_90[0]
activation_90 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_88 (Conv2D)	(None, None, None, 192)	208,896	block17_3_a
conv2d_91 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_88[0]
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_91[0]
activation_88 (Activation)	(None, None, None, 192)	0	batch_norma
activation_91 (Activation)	(None, None, None, 192)	0	batch_norma
block17_4_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_

block17_4_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_4_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_3_a block17_4_c
block17_4_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_93 (Conv2D)	(None, None, None, 128)	139,264	block17_4_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_93[C
activation_93 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_94 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_94[C
activation_94 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_92 (Conv2D)	(None, None, None, 192)	208,896	block17_4_a
conv2d_95 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_92[C
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_95[C
activation_92 (Activation)	(None, None, None, 192)	0	batch_norma
activation_95 (Activation)	(None, None, None, 192)	0	batch_norma
block17_5_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_5_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_5_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_4_a block17_5_c
block17_5_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_97 (Conv2D)	(None, None, None, 128)	139,264	block17_5_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_97[C
activation_97	(None, None,	0	batch_norma

(Activation)	None, 128)		
conv2d_98 (Conv2D)	(None, None, None, 160)	143,360	activation_98
batch_normalization_98 (BatchNormalization)	(None, None, None, 160)	480	conv2d_98[0]
activation_98 (Activation)	(None, None, None, 160)	0	batch_normalization_98
conv2d_96 (Conv2D)	(None, None, None, 192)	208,896	block17_5_activation_96
conv2d_99 (Conv2D)	(None, None, None, 192)	215,040	activation_99
batch_normalization_96 (BatchNormalization)	(None, None, None, 192)	576	conv2d_96[0]
batch_normalization_99 (BatchNormalization)	(None, None, None, 192)	576	conv2d_99[0]
activation_96 (Activation)	(None, None, None, 192)	0	batch_normalization_96
activation_99 (Activation)	(None, None, None, 192)	0	batch_normalization_99
block17_6_mixed (Concatenate)	(None, None, None, 384)	0	activation_96 activation_99
block17_6_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_6_mixed
custom_scale_layer_1 (CustomScaleLayer)	(None, None, None, 1088)	0	block17_5_activation_96 block17_6_conv
block17_6_activation (Activation)	(None, None, None, 1088)	0	custom_scale_layer_1
conv2d_101 (Conv2D)	(None, None, None, 128)	139,264	block17_6_activation
batch_normalization_101 (BatchNormalization)	(None, None, None, 128)	384	conv2d_101[0]
activation_101 (Activation)	(None, None, None, 128)	0	batch_normalization_101
conv2d_102 (Conv2D)	(None, None, None, 160)	143,360	activation_101
batch_normalization_102 (BatchNormalization)	(None, None, None, 160)	480	conv2d_102[0]
activation_102 (Activation)	(None, None, None, 160)	0	batch_normalization_102
conv2d_100 (Conv2D)	(None, None, None, 192)	208,896	block17_6_activation
conv2d_103 (Conv2D)	(None, None, None, 192)	215,040	activation_102

batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_100
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_103
activation_100 (Activation)	(None, None, None, 192)	0	batch_norma
activation_103 (Activation)	(None, None, None, 192)	0	batch_norma
block17_7_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_7_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_7_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_6_a block17_7_c
block17_7_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_105 (Conv2D)	(None, None, None, 128)	139,264	block17_7_a
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_105
activation_105 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_106 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_106
activation_106 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_104 (Conv2D)	(None, None, None, 192)	208,896	block17_7_a
conv2d_107 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_104
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_107
activation_104 (Activation)	(None, None, None, 192)	0	batch_norma
activation_107 (Activation)	(None, None, None, 192)	0	batch_norma
block17_8_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_

block17_8_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_8_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_7_α block17_8_α
block17_8_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_109 (Conv2D)	(None, None, None, 128)	139,264	block17_8_α
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_109
activation_109 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_110 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_110
activation_110 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_108 (Conv2D)	(None, None, None, 192)	208,896	block17_8_α
conv2d_111 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_108
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_111
activation_108 (Activation)	(None, None, None, 192)	0	batch_norma
activation_111 (Activation)	(None, None, None, 192)	0	batch_norma
block17_9_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_9_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_9_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_8_α block17_9_α
block17_9_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_113 (Conv2D)	(None, None, None, 128)	139,264	block17_9_α
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_113
activation_113	(None, None,	0	batch_norma

(Activation)	None, 128)		
conv2d_114 (Conv2D)	(None, None, None, 160)	143,360	activation_114
batch_normalization_114 (BatchNormalization)	(None, None, None, 160)	480	conv2d_114
activation_114 (Activation)	(None, None, None, 160)	0	batch_normalization_114
conv2d_112 (Conv2D)	(None, None, None, 192)	208,896	block17_9_activation
conv2d_115 (Conv2D)	(None, None, None, 192)	215,040	activation_112
batch_normalization_112 (BatchNormalization)	(None, None, None, 192)	576	conv2d_112
batch_normalization_115 (BatchNormalization)	(None, None, None, 192)	576	conv2d_115
activation_112 (Activation)	(None, None, None, 192)	0	batch_normalization_112
activation_115 (Activation)	(None, None, None, 192)	0	batch_normalization_115
block17_10_mixed (Concatenate)	(None, None, None, 384)	0	activation_115
block17_10_conv (Conv2D)	(None, None, None, 1088)	418,880	activation_112
custom_scale_layer_10 (CustomScaleLayer)	(None, None, None, 1088)	0	block17_10_conv
block17_10_activation (Activation)	(None, None, None, 1088)	0	custom_scale_layer_10
conv2d_117 (Conv2D)	(None, None, None, 128)	139,264	block17_10_activation
batch_normalization_117 (BatchNormalization)	(None, None, None, 128)	384	conv2d_117
activation_117 (Activation)	(None, None, None, 128)	0	batch_normalization_117
conv2d_118 (Conv2D)	(None, None, None, 160)	143,360	activation_117
batch_normalization_118 (BatchNormalization)	(None, None, None, 160)	480	conv2d_118
activation_118 (Activation)	(None, None, None, 160)	0	batch_normalization_118
conv2d_116 (Conv2D)	(None, None, None, 192)	208,896	block17_10_mixed
conv2d_119 (Conv2D)	(None, None, None, 192)	215,040	conv2d_116

batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_116
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_119
activation_116 (Activation)	(None, None, None, 192)	0	batch_norma
activation_119 (Activation)	(None, None, None, 192)	0	batch_norma
block17_11_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_11_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_11_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_10_ block17_11_
block17_11_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_121 (Conv2D)	(None, None, None, 128)	139,264	block17_11_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_121
activation_121 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_122 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_122
activation_122 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_120 (Conv2D)	(None, None, None, 192)	208,896	block17_11_
conv2d_123 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_120
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_123
activation_120 (Activation)	(None, None, None, 192)	0	batch_norma
activation_123 (Activation)	(None, None, None, 192)	0	batch_norma
block17_12_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_

block17_12_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_12_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_11_ block17_12_
block17_12_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_125 (Conv2D)	(None, None, None, 128)	139,264	block17_12_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_125
activation_125 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_126 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_126
activation_126 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_124 (Conv2D)	(None, None, None, 192)	208,896	block17_12_
conv2d_127 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_124
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_127
activation_124 (Activation)	(None, None, None, 192)	0	batch_norma
activation_127 (Activation)	(None, None, None, 192)	0	batch_norma
block17_13_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_13_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_13_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_12_ block17_13_
block17_13_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_129 (Conv2D)	(None, None, None, 128)	139,264	block17_13_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_129
activation_129	(None, None,	0	batch_norma

(Activation)	None, 128)		
conv2d_130 (Conv2D)	(None, None, None, 160)	143,360	activation_130
batch_normalization_130 (BatchNormalization)	(None, None, None, 160)	480	conv2d_130
activation_130 (Activation)	(None, None, None, 160)	0	batch_normalization_130
conv2d_128 (Conv2D)	(None, None, None, 192)	208,896	block17_13
conv2d_131 (Conv2D)	(None, None, None, 192)	215,040	activation_128
batch_normalization_128 (BatchNormalization)	(None, None, None, 192)	576	conv2d_128
batch_normalization_131 (BatchNormalization)	(None, None, None, 192)	576	conv2d_131
activation_128 (Activation)	(None, None, None, 192)	0	batch_normalization_128
activation_131 (Activation)	(None, None, None, 192)	0	batch_normalization_131
block17_14_mixed (Concatenate)	(None, None, None, 384)	0	activation_131
block17_14_conv (Conv2D)	(None, None, None, 1088)	418,880	activation_128
custom_scale_layer_14 (CustomScaleLayer)	(None, None, None, 1088)	0	block17_14_mixed
block17_14_activation (Activation)	(None, None, None, 1088)	0	block17_14_conv
conv2d_133 (Conv2D)	(None, None, None, 128)	139,264	custom_scale_layer_14
batch_normalization_133 (BatchNormalization)	(None, None, None, 128)	384	block17_14_activation
activation_133 (Activation)	(None, None, None, 128)	0	conv2d_133
conv2d_134 (Conv2D)	(None, None, None, 160)	143,360	activation_133
batch_normalization_134 (BatchNormalization)	(None, None, None, 160)	480	conv2d_134
activation_134 (Activation)	(None, None, None, 160)	0	batch_normalization_134
conv2d_132 (Conv2D)	(None, None, None, 192)	208,896	activation_134
conv2d_135 (Conv2D)	(None, None, None, 192)	215,040	conv2d_132

batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_132
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_135
activation_132 (Activation)	(None, None, None, 192)	0	batch_norma
activation_135 (Activation)	(None, None, None, 192)	0	batch_norma
block17_15_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_15_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_15_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_14_ block17_15_
block17_15_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_137 (Conv2D)	(None, None, None, 128)	139,264	block17_15_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_137
activation_137 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_138 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_138
activation_138 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_136 (Conv2D)	(None, None, None, 192)	208,896	block17_15_
conv2d_139 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_136
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_139
activation_136 (Activation)	(None, None, None, 192)	0	batch_norma
activation_139 (Activation)	(None, None, None, 192)	0	batch_norma
block17_16_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_

block17_16_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_16_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_15_ block17_16_
block17_16_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_141 (Conv2D)	(None, None, None, 128)	139,264	block17_16_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_141
activation_141 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_142 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_142
activation_142 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_140 (Conv2D)	(None, None, None, 192)	208,896	block17_16_
conv2d_143 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_140
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_143
activation_140 (Activation)	(None, None, None, 192)	0	batch_norma
activation_143 (Activation)	(None, None, None, 192)	0	batch_norma
block17_17_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_17_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_17_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_16_ block17_17_
block17_17_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_145 (Conv2D)	(None, None, None, 128)	139,264	block17_17_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_145
activation_145	(None, None,	0	batch_norma

(Activation)	None, 128)		
conv2d_146 (Conv2D)	(None, None, None, 160)	143,360	activation_146
batch_normalization_146 (BatchNormalization)	(None, None, None, 160)	480	conv2d_146
activation_146 (Activation)	(None, None, None, 160)	0	batch_normalization_146
conv2d_144 (Conv2D)	(None, None, None, 192)	208,896	block17_17
conv2d_147 (Conv2D)	(None, None, None, 192)	215,040	activation_144
batch_normalization_144 (BatchNormalization)	(None, None, None, 192)	576	conv2d_144
batch_normalization_147 (BatchNormalization)	(None, None, None, 192)	576	conv2d_147
activation_144 (Activation)	(None, None, None, 192)	0	batch_normalization_144
activation_147 (Activation)	(None, None, None, 192)	0	batch_normalization_147
block17_18_mixed (Concatenate)	(None, None, None, 384)	0	activation_147
block17_18_conv (Conv2D)	(None, None, None, 1088)	418,880	activation_144
custom_scale_layer_18 (CustomScaleLayer)	(None, None, None, 1088)	0	block17_18_conv
block17_18_activation (Activation)	(None, None, None, 1088)	0	custom_scale_layer_18
conv2d_149 (Conv2D)	(None, None, None, 128)	139,264	block17_18_activation
batch_normalization_149 (BatchNormalization)	(None, None, None, 128)	384	conv2d_149
activation_149 (Activation)	(None, None, None, 128)	0	batch_normalization_149
conv2d_150 (Conv2D)	(None, None, None, 160)	143,360	activation_149
batch_normalization_150 (BatchNormalization)	(None, None, None, 160)	480	conv2d_150
activation_150 (Activation)	(None, None, None, 160)	0	batch_normalization_150
conv2d_148 (Conv2D)	(None, None, None, 192)	208,896	conv2d_150
conv2d_151 (Conv2D)	(None, None, None, 192)	215,040	activation_150

batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_148
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_151
activation_148 (Activation)	(None, None, None, 192)	0	batch_norma
activation_151 (Activation)	(None, None, None, 192)	0	batch_norma
block17_19_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_
block17_19_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_19_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_18_ block17_19_
block17_19_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_153 (Conv2D)	(None, None, None, 128)	139,264	block17_19_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 128)	384	conv2d_153
activation_153 (Activation)	(None, None, None, 128)	0	batch_norma
conv2d_154 (Conv2D)	(None, None, None, 160)	143,360	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 160)	480	conv2d_154
activation_154 (Activation)	(None, None, None, 160)	0	batch_norma
conv2d_152 (Conv2D)	(None, None, None, 192)	208,896	block17_19_
conv2d_155 (Conv2D)	(None, None, None, 192)	215,040	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_152
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_155
activation_152 (Activation)	(None, None, None, 192)	0	batch_norma
activation_155 (Activation)	(None, None, None, 192)	0	batch_norma
block17_20_mixed (Concatenate)	(None, None, None, 384)	0	activation_ activation_

block17_20_conv (Conv2D)	(None, None, None, 1088)	418,880	block17_20_
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 1088)	0	block17_19_ block17_20_
block17_20_ac (Activation)	(None, None, None, 1088)	0	custom_sca1
conv2d_160 (Conv2D)	(None, None, None, 256)	278,528	block17_20_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_160
activation_160 (Activation)	(None, None, None, 256)	0	batch_norma
conv2d_156 (Conv2D)	(None, None, None, 256)	278,528	block17_20_
conv2d_158 (Conv2D)	(None, None, None, 256)	278,528	block17_20_
conv2d_161 (Conv2D)	(None, None, None, 288)	663,552	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_156
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_158
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 288)	864	conv2d_161
activation_156 (Activation)	(None, None, None, 256)	0	batch_norma
activation_158 (Activation)	(None, None, None, 256)	0	batch_norma
activation_161 (Activation)	(None, None, None, 288)	0	batch_norma
conv2d_157 (Conv2D)	(None, None, None, 384)	884,736	activation_
conv2d_159 (Conv2D)	(None, None, None, 288)	663,552	activation_
conv2d_162 (Conv2D)	(None, None, None, 320)	829,440	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 384)	1,152	conv2d_157
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 288)	864	conv2d_159
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 320)	960	conv2d_162
activation_157	(None, None,	0	batch_norma

(Activation)	None, 384)		
activation_159 (Activation)	(None, None, None, 288)	0	batch_norma
activation_162 (Activation)	(None, None, None, 320)	0	batch_norma
max_pooling2d_3 (MaxPooling2D)	(None, None, None, 1088)	0	block17_20_
mixed_7a (Concatenate)	(None, None, None, 2080)	0	activation_ activation_ activation_ max_pooling
conv2d_164 (Conv2D)	(None, None, None, 192)	399,360	mixed_7a[0.]
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_164
activation_164 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_165 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_165
activation_165 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_163 (Conv2D)	(None, None, None, 192)	399,360	mixed_7a[0.]
conv2d_166 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_163
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_166
activation_163 (Activation)	(None, None, None, 192)	0	batch_norma
activation_166 (Activation)	(None, None, None, 256)	0	batch_norma
block8_1_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_1_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_1_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	mixed_7a[0.] block8_1_cc
block8_1_ac (Activation)	(None, None, None, 2080)	0	custom_sca1

conv2d_168 (Conv2D)	(None, None, None, 192)	399,360	block8_1_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_168
activation_168 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_169 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_169
activation_169 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_167 (Conv2D)	(None, None, None, 192)	399,360	block8_1_ac
conv2d_170 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_167
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_170
activation_167 (Activation)	(None, None, None, 192)	0	batch_norma
activation_170 (Activation)	(None, None, None, 256)	0	batch_norma
block8_2_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_2_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_2_m:
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_1_ac block8_2_cc
block8_2_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_172 (Conv2D)	(None, None, None, 192)	399,360	block8_2_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_172
activation_172 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_173 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_173
activation_173	(None, None,	0	batch_norma

(Activation)	None, 224)		
conv2d_171 (Conv2D)	(None, None, None, 192)	399,360	block8_2_ac
conv2d_174 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_171
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_174
activation_171 (Activation)	(None, None, None, 192)	0	batch_norma
activation_174 (Activation)	(None, None, None, 256)	0	batch_norma
block8_3_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_3_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_3_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_2_ac block8_3_cc
block8_3_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_176 (Conv2D)	(None, None, None, 192)	399,360	block8_3_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_176
activation_176 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_177 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_177
activation_177 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_175 (Conv2D)	(None, None, None, 192)	399,360	block8_3_ac
conv2d_178 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_175
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_178
activation_175 (Activation)	(None, None, None, 192)	0	batch_norma

activation_178 (Activation)	(None, None, None, 256)	0	batch_norma
block8_4_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_4_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_4_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_3_ac block8_4_cc
block8_4_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_180 (Conv2D)	(None, None, None, 192)	399,360	block8_4_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_180
activation_180 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_181 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_181
activation_181 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_179 (Conv2D)	(None, None, None, 192)	399,360	block8_4_ac
conv2d_182 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_179
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_182
activation_179 (Activation)	(None, None, None, 192)	0	batch_norma
activation_182 (Activation)	(None, None, None, 256)	0	batch_norma
block8_5_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_5_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_5_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_4_ac block8_5_cc
block8_5_ac (Activation)	(None, None, None, 2080)	0	custom_sca1

conv2d_184 (Conv2D)	(None, None, None, 192)	399,360	block8_5_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_184
activation_184 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_185 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_185
activation_185 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_183 (Conv2D)	(None, None, None, 192)	399,360	block8_5_ac
conv2d_186 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_183
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_186
activation_183 (Activation)	(None, None, None, 192)	0	batch_norma
activation_186 (Activation)	(None, None, None, 256)	0	batch_norma
block8_6_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_6_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_6_m:
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_5_ac block8_6_cc
block8_6_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_188 (Conv2D)	(None, None, None, 192)	399,360	block8_6_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_188
activation_188 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_189 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_189
activation_189	(None, None,	0	batch_norma

(Activation)	None, 224)		
conv2d_187 (Conv2D)	(None, None, None, 192)	399,360	block8_6_ac
conv2d_190 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_187
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_190
activation_187 (Activation)	(None, None, None, 192)	0	batch_norma
activation_190 (Activation)	(None, None, None, 256)	0	batch_norma
block8_7_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_7_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_7_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_6_ac block8_7_cc
block8_7_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_192 (Conv2D)	(None, None, None, 192)	399,360	block8_7_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_192
activation_192 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_193 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_193
activation_193 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_191 (Conv2D)	(None, None, None, 192)	399,360	block8_7_ac
conv2d_194 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_191
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_194
activation_191 (Activation)	(None, None, None, 192)	0	batch_norma

activation_194 (Activation)	(None, None, None, 256)	0	batch_norma
block8_8_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_8_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_8_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_7_ac block8_8_cc
block8_8_ac (Activation)	(None, None, None, 2080)	0	custom_sca1
conv2d_196 (Conv2D)	(None, None, None, 192)	399,360	block8_8_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_196
activation_196 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_197 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_197
activation_197 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_195 (Conv2D)	(None, None, None, 192)	399,360	block8_8_ac
conv2d_198 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_195
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_198
activation_195 (Activation)	(None, None, None, 192)	0	batch_norma
activation_198 (Activation)	(None, None, None, 256)	0	batch_norma
block8_9_mixed (Concatenate)	(None, None, None, 448)	0	activation_ activation_
block8_9_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_9_mi
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_8_ac block8_9_cc
block8_9_ac (Activation)	(None, None, None, 2080)	0	custom_sca1

conv2d_200 (Conv2D)	(None, None, None, 192)	399,360	block8_9_ac
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_200
activation_200 (Activation)	(None, None, None, 192)	0	batch_norma
conv2d_201 (Conv2D)	(None, None, None, 224)	129,024	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 224)	672	conv2d_201
activation_201 (Activation)	(None, None, None, 224)	0	batch_norma
conv2d_199 (Conv2D)	(None, None, None, 192)	399,360	block8_9_ac
conv2d_202 (Conv2D)	(None, None, None, 256)	172,032	activation_
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 192)	576	conv2d_199
batch_normalizatio... (BatchNormalizatio...	(None, None, None, 256)	768	conv2d_202
activation_199 (Activation)	(None, None, None, 192)	0	batch_norma
activation_202 (Activation)	(None, None, None, 256)	0	batch_norma
block8_10_mixed (Concatenate)	(None, None, None, 448)	0	activation_
block8_10_conv (Conv2D)	(None, None, None, 2080)	933,920	block8_10_r
custom_scale_layer... (CustomScaleLayer)	(None, None, None, 2080)	0	block8_9_ac
conv_7b (Conv2D)	(None, None, None, 1536)	3,194,880	custom_sca1
conv_7b_bn (BatchNormalizatio...	(None, None, None, 1536)	4,608	conv_7b[0]
conv_7b_ac (Activation)	(None, None, None, 1536)	0	conv_7b_bn

Total params: 54,336,736 (207.28 MB)

Trainable params: 54,276,192 (207.05 MB)

Non-trainable params: 60,544 (236.50 KB)

```
In [7]: data_augmentation = keras.Sequential(
        [
```

```

        layers.RandomFlip("horizontal"),
        layers.RandomRotation(0.1),
        layers.RandomZoom(0.2),
    ]
)

inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)
x = keras.applications.inception_resnet_v2.preprocess_input(x)
x = conv_base(x)
x = layers.Flatten()(x)
x = layers.Dense(256)(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])

```

```

In [8]: callbacks = [
        keras.callbacks.ModelCheckpoint(
            filepath="inception.keras",
            save_best_only=True,
            monitor="val_loss")
    ]
history = model.fit(
    train_dataset,
    epochs=100,
    validation_data=validation_dataset,
    callbacks=callbacks)

```

Epoch 1/100
63/63 ————— 126s 2s/step - accuracy: 0.5315 - loss: 18.2568 - val_accuracy: 0.5360 - val_loss: 0.7795
Epoch 2/100
63/63 ————— 100s 2s/step - accuracy: 0.4818 - loss: 10.1322 - val_accuracy: 0.5160 - val_loss: 0.7143
Epoch 3/100
63/63 ————— 99s 2s/step - accuracy: 0.5012 - loss: 3.2025 - val_accuracy: 0.5010 - val_loss: 0.619263
Epoch 4/100
63/63 ————— 102s 2s/step - accuracy: 0.4923 - loss: 3.3084 - val_accuracy: 0.5020 - val_loss: 0.6982
Epoch 5/100
63/63 ————— 99s 2s/step - accuracy: 0.5311 - loss: 1.0101 - val_accuracy: 0.5220 - val_loss: 0.7050
Epoch 6/100
63/63 ————— 103s 2s/step - accuracy: 0.5607 - loss: 0.7491 - val_accuracy: 0.5000 - val_loss: 0.7084
Epoch 7/100
63/63 ————— 102s 2s/step - accuracy: 0.6020 - loss: 0.6889 - val_accuracy: 0.6120 - val_loss: 0.6633
Epoch 8/100
63/63 ————— 101s 2s/step - accuracy: 0.6641 - loss: 0.6349 - val_accuracy: 0.6170 - val_loss: 0.6589
Epoch 9/100
63/63 ————— 100s 2s/step - accuracy: 0.6836 - loss: 0.6106 - val_accuracy: 0.6390 - val_loss: 0.6770
Epoch 10/100
63/63 ————— 99s 2s/step - accuracy: 0.6994 - loss: 0.5994 - val_accuracy: 0.6210 - val_loss: 0.6697
Epoch 11/100
63/63 ————— 102s 2s/step - accuracy: 0.7123 - loss: 0.5693 - val_accuracy: 0.6620 - val_loss: 0.6295
Epoch 12/100
63/63 ————— 101s 2s/step - accuracy: 0.7185 - loss: 0.5469 - val_accuracy: 0.6470 - val_loss: 0.6303
Epoch 13/100
63/63 ————— 102s 2s/step - accuracy: 0.7095 - loss: 0.5460 - val_accuracy: 0.6650 - val_loss: 0.6100
Epoch 14/100
63/63 ————— 100s 2s/step - accuracy: 0.7277 - loss: 0.5371 - val_accuracy: 0.6410 - val_loss: 0.6556
Epoch 15/100
63/63 ————— 99s 2s/step - accuracy: 0.7379 - loss: 0.5155 - val_accuracy: 0.6350 - val_loss: 0.7237
Epoch 16/100
63/63 ————— 101s 2s/step - accuracy: 0.7105 - loss: 0.5662 - val_accuracy: 0.5980 - val_loss: 0.6608
Epoch 17/100
63/63 ————— 103s 2s/step - accuracy: 0.7262 - loss: 0.5369 - val_accuracy: 0.6950 - val_loss: 0.5581
Epoch 18/100
63/63 ————— 101s 2s/step - accuracy: 0.7426 - loss: 0.5282 - val_accuracy: 0.6360 - val_loss: 0.7468
Epoch 19/100
63/63 ————— 100s 2s/step - accuracy: 0.7466 - loss: 0.5226 - val_accuracy: 0.6920 - val_loss: 0.6073
Epoch 20/100
63/63 ————— 100s 2s/step - accuracy: 0.7588 - loss: 0.4979 - val_accuracy: 0.7070 - val_loss: 0.5828

Epoch 21/100
63/63  102s 2s/step - accuracy: 0.7675 - loss: 0.4688 - val_accuracy: 0.7420 - val_loss: 0.5287
Epoch 22/100
63/63  101s 2s/step - accuracy: 0.7772 - loss: 0.4728 - val_accuracy: 0.7390 - val_loss: 0.5366
Epoch 23/100
63/63  102s 2s/step - accuracy: 0.7743 - loss: 0.4766 - val_accuracy: 0.7500 - val_loss: 0.4938
Epoch 24/100
63/63  100s 2s/step - accuracy: 0.7746 - loss: 0.4680 - val_accuracy: 0.6630 - val_loss: 0.6141
Epoch 25/100
63/63  101s 2s/step - accuracy: 0.7809 - loss: 0.4681 - val_accuracy: 0.5780 - val_loss: 1.0521
Epoch 26/100
63/63  100s 2s/step - accuracy: 0.7791 - loss: 0.4561 - val_accuracy: 0.7410 - val_loss: 0.5639
Epoch 27/100
63/63  102s 2s/step - accuracy: 0.7925 - loss: 0.4347 - val_accuracy: 0.7650 - val_loss: 0.4865
Epoch 28/100
63/63  100s 2s/step - accuracy: 0.8152 - loss: 0.4033 - val_accuracy: 0.6840 - val_loss: 0.6886
Epoch 29/100
63/63  101s 2s/step - accuracy: 0.8153 - loss: 0.4073 - val_accuracy: 0.7610 - val_loss: 0.4675
Epoch 30/100
63/63  101s 2s/step - accuracy: 0.8282 - loss: 0.4097 - val_accuracy: 0.7200 - val_loss: 0.6000
Epoch 31/100
63/63  100s 2s/step - accuracy: 0.7947 - loss: 0.4051 - val_accuracy: 0.7540 - val_loss: 0.5999
Epoch 32/100
63/63  101s 2s/step - accuracy: 0.8293 - loss: 0.3879 - val_accuracy: 0.7750 - val_loss: 0.5146
Epoch 33/100
63/63  100s 2s/step - accuracy: 0.8266 - loss: 0.3867 - val_accuracy: 0.7500 - val_loss: 0.5765
Epoch 34/100
63/63  100s 2s/step - accuracy: 0.8396 - loss: 0.3676 - val_accuracy: 0.8380 - val_loss: 0.3832
Epoch 35/100
63/63  100s 2s/step - accuracy: 0.8398 - loss: 0.3647 - val_accuracy: 0.7760 - val_loss: 0.4578
Epoch 36/100
63/63  101s 2s/step - accuracy: 0.8407 - loss: 0.3666 - val_accuracy: 0.8060 - val_loss: 0.4668
Epoch 37/100
63/63  101s 2s/step - accuracy: 0.8456 - loss: 0.3563 - val_accuracy: 0.7820 - val_loss: 0.5779
Epoch 38/100
63/63  99s 2s/step - accuracy: 0.8342 - loss: 0.3628 - val_accuracy: 0.7550 - val_loss: 0.6625
Epoch 39/100
63/63  99s 2s/step - accuracy: 0.8578 - loss: 0.3271 - val_accuracy: 0.8140 - val_loss: 0.5162
Epoch 40/100
63/63  99s 2s/step - accuracy: 0.8721 - loss: 0.3063 - val_accuracy: 0.7800 - val_loss: 0.5349

Epoch 41/100
63/63  101s 2s/step - accuracy: 0.8546 - loss: 0.3194 - val_accuracy: 0.7450 - val_loss: 0.6807
Epoch 42/100
63/63  100s 2s/step - accuracy: 0.8642 - loss: 0.3062 - val_accuracy: 0.7350 - val_loss: 0.6244
Epoch 43/100
63/63  102s 2s/step - accuracy: 0.8841 - loss: 0.2830 - val_accuracy: 0.8580 - val_loss: 0.3432
Epoch 44/100
63/63  99s 2s/step - accuracy: 0.8812 - loss: 0.3038 - val_accuracy: 0.8110 - val_loss: 0.4804
Epoch 45/100
63/63  99s 2s/step - accuracy: 0.8617 - loss: 0.3087 - val_accuracy: 0.8240 - val_loss: 0.4475
Epoch 46/100
63/63  100s 2s/step - accuracy: 0.8929 - loss: 0.2611 - val_accuracy: 0.8380 - val_loss: 0.4060
Epoch 47/100
63/63  101s 2s/step - accuracy: 0.8992 - loss: 0.2326 - val_accuracy: 0.5580 - val_loss: 2.2986
Epoch 48/100
63/63  100s 2s/step - accuracy: 0.8848 - loss: 0.3061 - val_accuracy: 0.7810 - val_loss: 0.4953
Epoch 49/100
63/63  100s 2s/step - accuracy: 0.9113 - loss: 0.2306 - val_accuracy: 0.8120 - val_loss: 0.4880
Epoch 50/100
63/63  100s 2s/step - accuracy: 0.9000 - loss: 0.2317 - val_accuracy: 0.8230 - val_loss: 0.6186
Epoch 51/100
63/63  100s 2s/step - accuracy: 0.9073 - loss: 0.2304 - val_accuracy: 0.8260 - val_loss: 0.4587
Epoch 52/100
63/63  101s 2s/step - accuracy: 0.9095 - loss: 0.2103 - val_accuracy: 0.8310 - val_loss: 0.4118
Epoch 53/100
63/63  101s 2s/step - accuracy: 0.9177 - loss: 0.2096 - val_accuracy: 0.8090 - val_loss: 0.5906
Epoch 54/100
63/63  98s 2s/step - accuracy: 0.9083 - loss: 0.2986 - val_accuracy: 0.8120 - val_loss: 0.6363
Epoch 55/100
63/63  100s 2s/step - accuracy: 0.9097 - loss: 0.2166 - val_accuracy: 0.8570 - val_loss: 0.4926
Epoch 56/100
63/63  99s 2s/step - accuracy: 0.9240 - loss: 0.1806 - val_accuracy: 0.8720 - val_loss: 0.3758
Epoch 57/100
63/63  100s 2s/step - accuracy: 0.9227 - loss: 0.1945 - val_accuracy: 0.7700 - val_loss: 1.0870
Epoch 58/100
63/63  99s 2s/step - accuracy: 0.9365 - loss: 0.1869 - val_accuracy: 0.8390 - val_loss: 0.4162
Epoch 59/100
63/63  100s 2s/step - accuracy: 0.9307 - loss: 0.1645 - val_accuracy: 0.8260 - val_loss: 0.4518
Epoch 60/100
63/63  101s 2s/step - accuracy: 0.9378 - loss: 0.1803 - val_accuracy: 0.8070 - val_loss: 0.7821

Epoch 61/100
63/63  **99s** 2s/step - accuracy: 0.9330 - loss: 0.1692 - val_accuracy: 0.8620 - val_loss: 0.4174
Epoch 62/100
63/63  **99s** 2s/step - accuracy: 0.9451 - loss: 0.1438 - val_accuracy: 0.8730 - val_loss: 0.3465
Epoch 63/100
63/63  **100s** 2s/step - accuracy: 0.9295 - loss: 0.1829 - val_accuracy: 0.8360 - val_loss: 0.4757
Epoch 64/100
63/63  **101s** 2s/step - accuracy: 0.9386 - loss: 0.1652 - val_accuracy: 0.8690 - val_loss: 0.3432
Epoch 65/100
63/63  **100s** 2s/step - accuracy: 0.9448 - loss: 0.1358 - val_accuracy: 0.8420 - val_loss: 0.5862
Epoch 66/100
63/63  **98s** 2s/step - accuracy: 0.9400 - loss: 0.1542 - val_accuracy: 0.8610 - val_loss: 0.3757
Epoch 67/100
63/63  **99s** 2s/step - accuracy: 0.9393 - loss: 0.1420 - val_accuracy: 0.8620 - val_loss: 0.4023
Epoch 68/100
63/63  **99s** 2s/step - accuracy: 0.9529 - loss: 0.1325 - val_accuracy: 0.8520 - val_loss: 0.4314
Epoch 69/100
63/63  **100s** 2s/step - accuracy: 0.9512 - loss: 0.1351 - val_accuracy: 0.8650 - val_loss: 0.4155
Epoch 70/100
63/63  **100s** 2s/step - accuracy: 0.9506 - loss: 0.1419 - val_accuracy: 0.8750 - val_loss: 0.4041
Epoch 71/100
63/63  **99s** 2s/step - accuracy: 0.9438 - loss: 0.1457 - val_accuracy: 0.8690 - val_loss: 0.3919
Epoch 72/100
63/63  **100s** 2s/step - accuracy: 0.9596 - loss: 0.1262 - val_accuracy: 0.8310 - val_loss: 0.5697
Epoch 73/100
63/63  **101s** 2s/step - accuracy: 0.9410 - loss: 0.1677 - val_accuracy: 0.8550 - val_loss: 0.3918
Epoch 74/100
63/63  **99s** 2s/step - accuracy: 0.9546 - loss: 0.1285 - val_accuracy: 0.8390 - val_loss: 0.5456
Epoch 75/100
63/63  **99s** 2s/step - accuracy: 0.9585 - loss: 0.1118 - val_accuracy: 0.8030 - val_loss: 0.7654
Epoch 76/100
63/63  **100s** 2s/step - accuracy: 0.9645 - loss: 0.0966 - val_accuracy: 0.8650 - val_loss: 0.4339
Epoch 77/100
63/63  **99s** 2s/step - accuracy: 0.9559 - loss: 0.1303 - val_accuracy: 0.8280 - val_loss: 0.6432
Epoch 78/100
63/63  **98s** 2s/step - accuracy: 0.9546 - loss: 0.1060 - val_accuracy: 0.8650 - val_loss: 0.5717
Epoch 79/100
63/63  **98s** 2s/step - accuracy: 0.9657 - loss: 0.0919 - val_accuracy: 0.8450 - val_loss: 0.5124
Epoch 80/100
63/63  **100s** 2s/step - accuracy: 0.9680 - loss: 0.1191 - val_accuracy: 0.8830 - val_loss: 0.3931

Epoch 81/100
63/63  100s 2s/step - accuracy: 0.9580 - loss: 0.1117 - val_accuracy: 0.8540 - val_loss: 0.5043
Epoch 82/100
63/63  99s 2s/step - accuracy: 0.9655 - loss: 0.0998 - val_accuracy: 0.8480 - val_loss: 0.5440
Epoch 83/100
63/63  98s 2s/step - accuracy: 0.9622 - loss: 0.0890 - val_accuracy: 0.8280 - val_loss: 0.5124
Epoch 84/100
63/63  100s 2s/step - accuracy: 0.9679 - loss: 0.0948 - val_accuracy: 0.8690 - val_loss: 0.4482
Epoch 85/100
63/63  100s 2s/step - accuracy: 0.9716 - loss: 0.0743 - val_accuracy: 0.8640 - val_loss: 0.3957
Epoch 86/100
63/63  100s 2s/step - accuracy: 0.9710 - loss: 0.0746 - val_accuracy: 0.8670 - val_loss: 0.4356
Epoch 87/100
63/63  99s 2s/step - accuracy: 0.9782 - loss: 0.0690 - val_accuracy: 0.8790 - val_loss: 0.4410
Epoch 88/100
63/63  99s 2s/step - accuracy: 0.9808 - loss: 0.0584 - val_accuracy: 0.8580 - val_loss: 0.4863
Epoch 89/100
63/63  99s 2s/step - accuracy: 0.9761 - loss: 0.0655 - val_accuracy: 0.8040 - val_loss: 0.8243
Epoch 90/100
63/63  100s 2s/step - accuracy: 0.9635 - loss: 0.0938 - val_accuracy: 0.8700 - val_loss: 0.4156
Epoch 91/100
63/63  100s 2s/step - accuracy: 0.9776 - loss: 0.0683 - val_accuracy: 0.8770 - val_loss: 0.4223
Epoch 92/100
63/63  99s 2s/step - accuracy: 0.9606 - loss: 0.1035 - val_accuracy: 0.8550 - val_loss: 0.5724
Epoch 93/100
63/63  100s 2s/step - accuracy: 0.9781 - loss: 0.0734 - val_accuracy: 0.8720 - val_loss: 0.5713
Epoch 94/100
63/63  100s 2s/step - accuracy: 0.9665 - loss: 0.0920 - val_accuracy: 0.8580 - val_loss: 0.5660
Epoch 95/100
63/63  100s 2s/step - accuracy: 0.9750 - loss: 0.0772 - val_accuracy: 0.8250 - val_loss: 0.7102
Epoch 96/100
63/63  98s 2s/step - accuracy: 0.9664 - loss: 0.0935 - val_accuracy: 0.8700 - val_loss: 0.6126
Epoch 97/100
63/63  100s 2s/step - accuracy: 0.9740 - loss: 0.0826 - val_accuracy: 0.8770 - val_loss: 0.4139
Epoch 98/100
63/63  101s 2s/step - accuracy: 0.9718 - loss: 0.0567 - val_accuracy: 0.8210 - val_loss: 0.7217
Epoch 99/100
63/63  99s 2s/step - accuracy: 0.9706 - loss: 0.0757 - val_accuracy: 0.8730 - val_loss: 0.4600
Epoch 100/100
63/63  99s 2s/step - accuracy: 0.9672 - loss: 0.0885 - val_accuracy: 0.8360 - val_loss: 0.7174

```
In [9]: import matplotlib.pyplot as plt
```

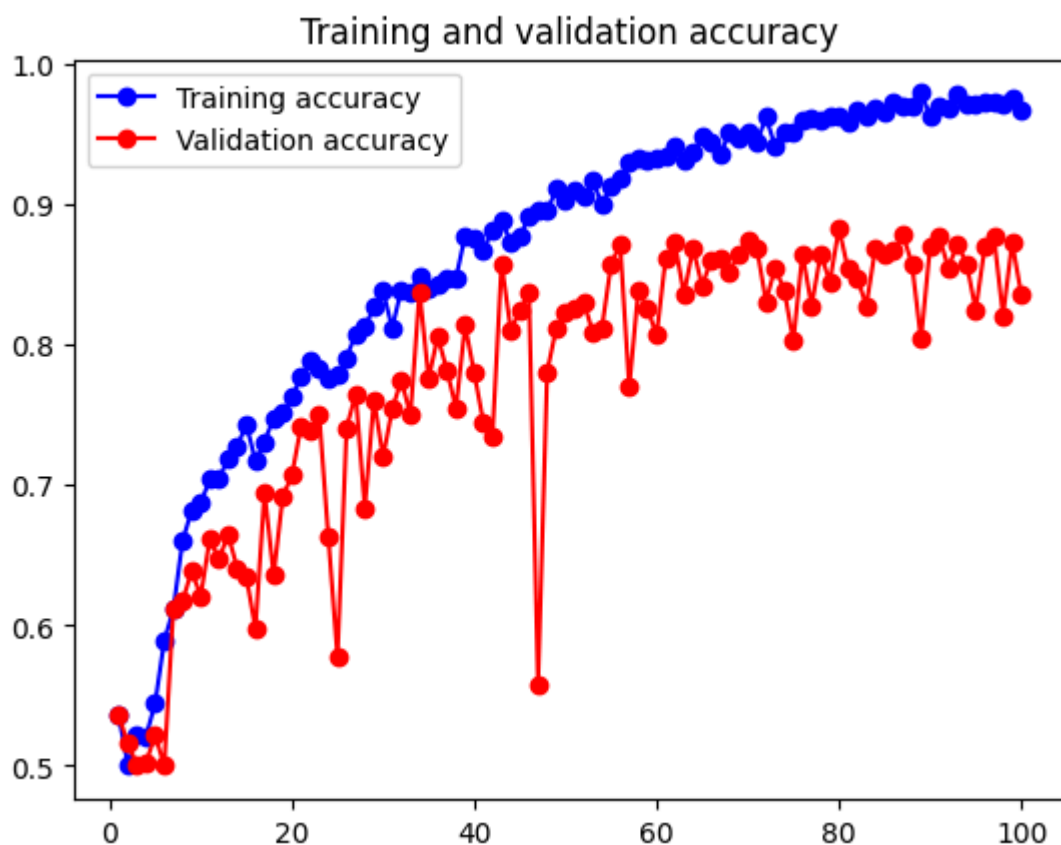
```
def plot(history):
    accuracy = history.history["accuracy"]
    val_accuracy = history.history["val_accuracy"]
    loss = history.history["loss"]
    val_loss = history.history["val_loss"]

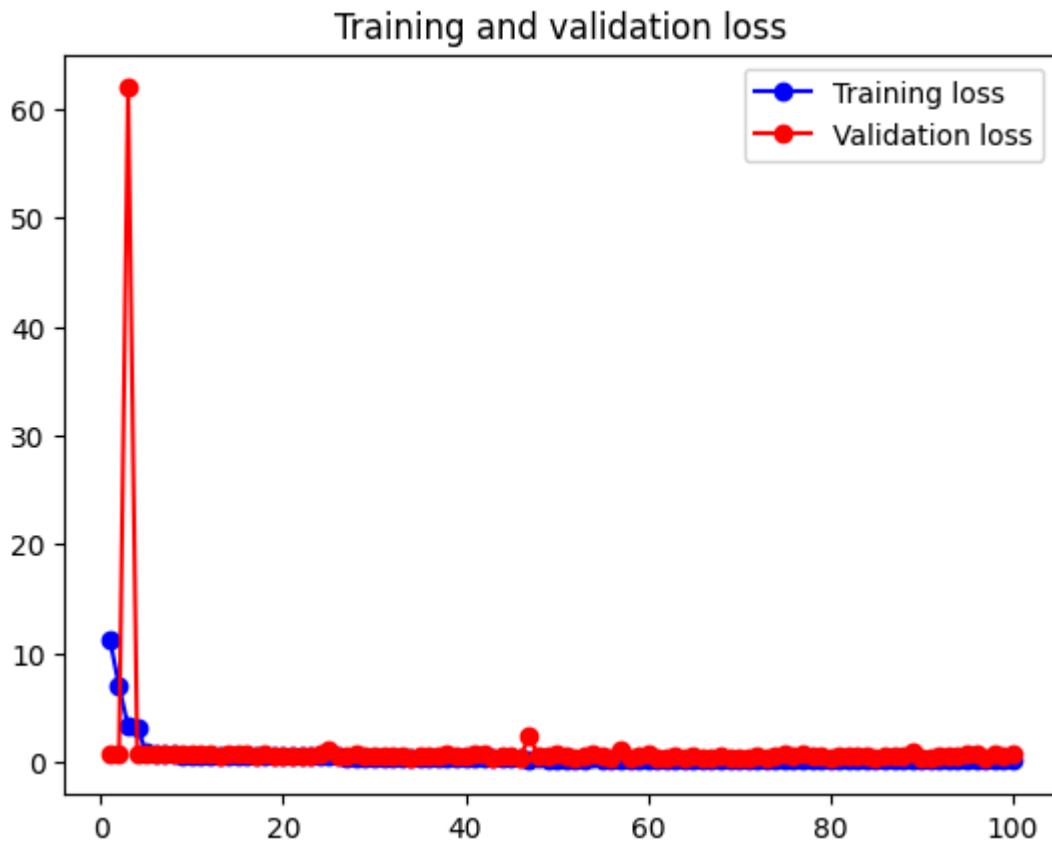
    epochs = range(1, len(accuracy) + 1)

    plt.plot(epochs, accuracy, "b-o", label="Training accuracy")
    plt.plot(epochs, val_accuracy, "r-o", label="Validation accuracy")
    plt.title("Training and validation accuracy")
    plt.legend()
    plt.figure()

    plt.plot(epochs, loss, "b-o", label="Training loss")
    plt.plot(epochs, val_loss, "r-o", label="Validation loss")
    plt.title("Training and validation loss")
    plt.legend()
    plt.show()
```

```
In [10]: plot(history)
```





```
In [11]: model.load_weights("inception.keras")
test_loss, test_acc = model.evaluate(test_dataset)
print(f"Test accuracy: {test_acc:.3f}")
```

63/63 ————— 21s 341ms/step - accuracy: 0.8667 - loss: 0.4041
Test accuracy: 0.850

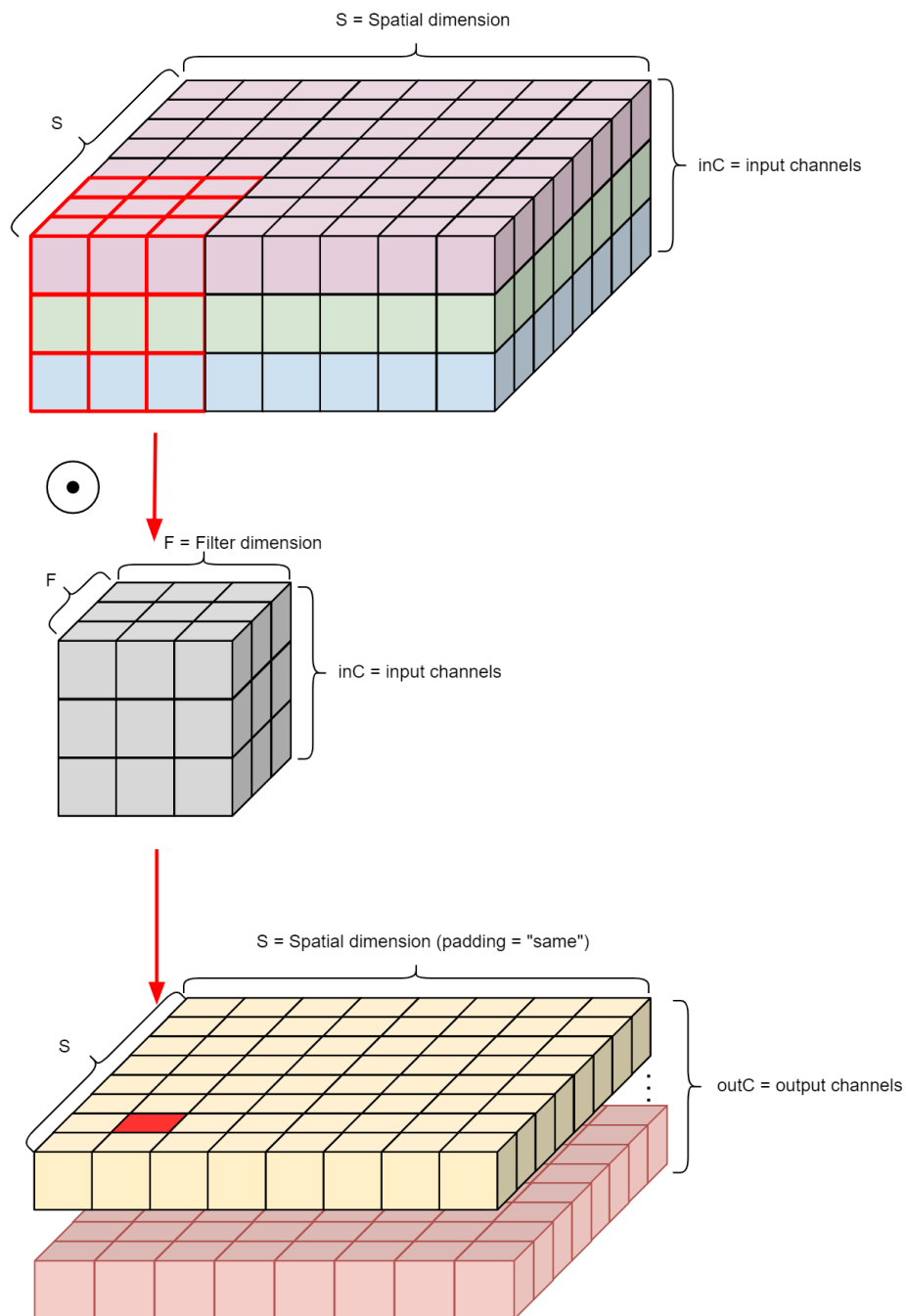
3.11. Depthwise separable convolutions (Xception networks)

- Source of images:
 - <https://eli.thegreenplace.net/2018/depthwise-separable-convolutions-for-machine-learning/>
 - <https://towardsdatascience.com/a-basic-introduction-to-separable-convolutions-b99ec3102728>
 - <https://towardsdatascience.com/a-comprehensive-introduction-to-different-types-of-convolutions-in-deep-learning-669281e58215>

3.11.1. Normal convolutions

- A convolution layer attempts to learn filters in a 3D space, with 2 spatial dimensions (width and height) and a channel dimension.
- Thus a single convolution kernel is tasked with simultaneously mapping cross-channel correlations and spatial correlations.
- The number of trainable parameters and the number of operations to perform the convolution are considerable (notably if the kernel dimension is high).
- Let's define the following **variables**:

- S : spatial dimension - width and height, assuming square inputs.
- F : filter width and height, assuming square filter.
- inC : number of input channels.
- $outC$: number of output channels.
- Padding
 - We also assume **same** padding, so that the spatial size of the output matches the input.
 - If we were not using padding then spatial size would shrink by two tiles alongside each dimension. So we would need to considerate as spatial dimensions $(S - 2) \times (S - 2)$.



- **Parameters = $F \times F \times inC \times outC$**
 - $(F \times F \times inC)$ is the dimension of a 3D filter

- `outC` is the number of filters
- **Operations** = $F \times F \times \text{inC} \times S \times S \times \text{outC}$
 - $(F \times F \times \text{inC})$ at each position of the spatial dimension
 - $(S \times S)$ multiplied by the size of the spatial dimension (input size)
 - `outC` multiplied by the number of filters (output channels)
- **Example:** $S = 128, F = 3, \text{inC} = 3, \text{outC} = 16$
 - **Parameters:** $3 \times 3 \times 3 \times 16 = 432$
 - **Operations:** $3 \times 3 \times 3 \times 128 \times 128 \times 16 = 7.077.888$

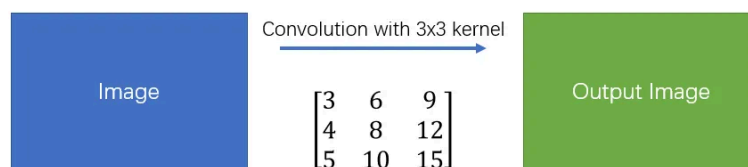
3.11.2. Spatial separable convolutions

- A spatial separable convolution simply divides a kernel into two, smaller kernels. The most common case would be to divide a 3x3 kernel into a 3x1 and 1x3 kernel, like so:

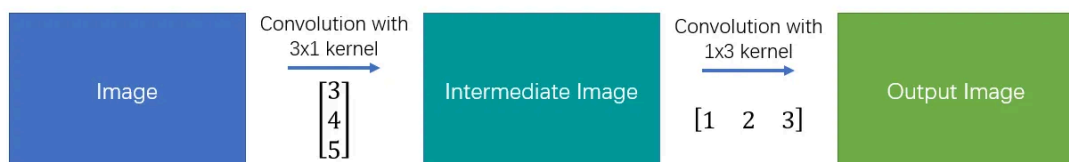
$$\begin{bmatrix} 3 & 6 & 9 \\ 4 & 8 & 12 \\ 5 & 10 & 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$$

- Now, instead of doing one convolution with 9 multiplications, we do two convolutions with 3 multiplications each (6 in total) to achieve the same effect.
- With less multiplications, computational complexity goes down, and the network is able to run faster.

Simple Convolution



Spatial Separable Convolution



- The main issue with the spatial separable convolution is that not all kernels can be "separated" into two, smaller kernels.
- This becomes particularly bothersome during training, if we replace all traditional convolutions by the spatially separable convolution, we limit ourselves for searching all possible kernels and the training results may be sub-optimal.

3.11.3. Depthwise separable convolutions

Inception architectures

- A normal convolutional kernel is tasked with simultaneously mapping cross-channel correlations and spatial correlations
- This idea behind the Inception module is to make this process easier and more efficient by explicitly factoring it into a series of operations that would independently look at cross-channel correlations (the 1x1 convolution step) and at spatial correlations (the normal convolution step).
- The fundamental hypothesis behind Inception is that **cross-channel correlations and spatial correlations are sufficiently decoupled that it is preferable not to map them jointly.**

Xception architectures

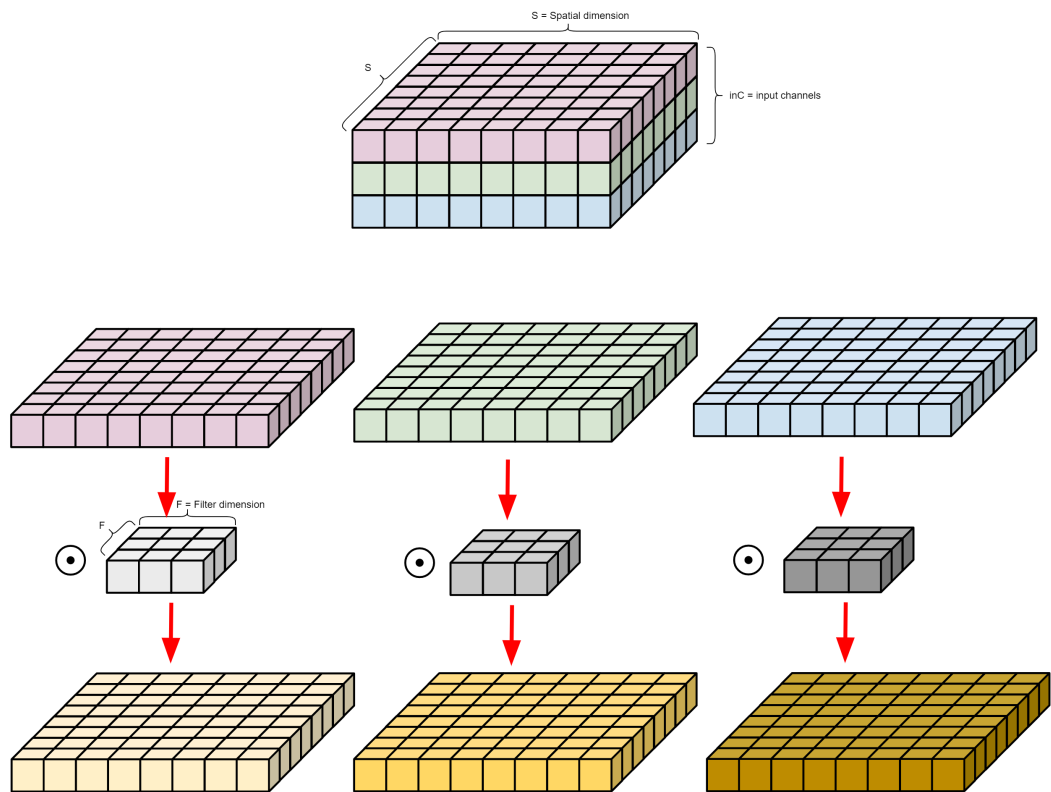
- The Xception architecture make the following hypothesis: that **the mapping of cross-channels correlations and spatial correlations in the feature maps of convolutional neural networks can be entirely decoupled.**
- Because this assumption is generally true for the image representations learned by deep neural networks, it serves as a useful prior that helps the model make more efficient use of its training data.
- Since this hypothesis is a stronger version of the hypothesis underlying the Inception architecture, the authors named their proposed architecture **Xception**, which stands for "*Extreme Inception*" (see the original paper at <https://arxiv.org/abs/1610.02357>).

Depthwise separable convolutions

- A depthwise separable convolution **splits a kernel into 2 separate kernels that do two convolutions: the depthwise convolution and the pointwise convolution.**
- Unlike spatial separable convolutions, depthwise separable convolutions work with kernels that cannot be "factored" into two smaller kernels. Hence, it is more commonly used.

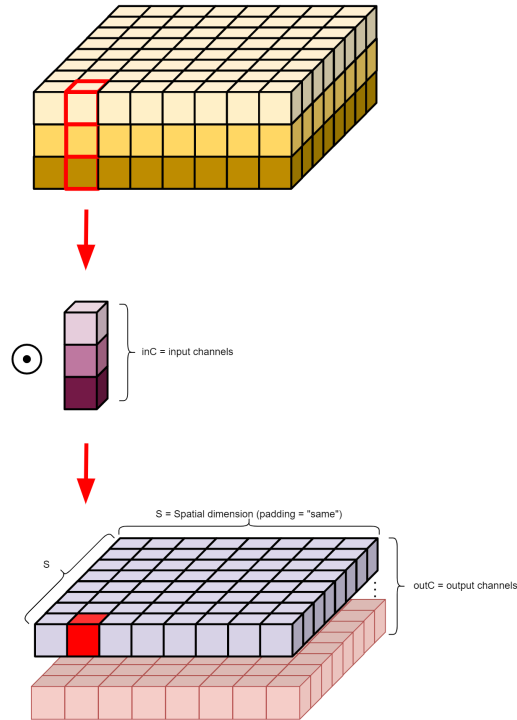
Depthwise convolution

- In the depthwise convolution, we give the input image a convolution without changing the depth.
- **It is a spatial convolution performed independently over each channel of an input.**
- **Parameters = $F \times F \times \text{inC}$**
 - $(F \times F)$ is the dimension of the filters used.
 - inC is the number of input channels.
- **Operations = $F \times F \times \text{inC} \times S \times S$**
 - $(F \times F \times \text{inC})$ at each position of the spatial dimension
 - $(S \times S)$ multiplied by the size of the spatial dimension (input size)



Pointwise convolution

- It is a 1×1 convolution, projecting the channels output by the depthwise convolution onto a new channel space.
- **Depthwise convolution calculation**
 - In the depthwise convolution, assuming a **same** padding, we have the same dimensions in the image ($S \times S$) and the same depth **inC**
 - Now, we need to increase the number of channels of the image with the pointwise convolution.
 - The pointwise convolution is so named because it uses a 1×1 kernel
 - This kernel has a depth of **inC**, the number of channels of the input image.
 - We apply **outC** kernels to obtain **outC** channels in the output image.
- **Parameters = $inC \times outC$**
 - **inC** is the number of input channels.
 - **outC** is the number of output channels.
- **Operations = $inC \times S \times S \times outC$**
 - **inC** times at each position of the spatial dimension
 - ($S \times S$) multiplied by the size of the spatial dimension (input size)
 - multiplied by the number of output channels **outC**.

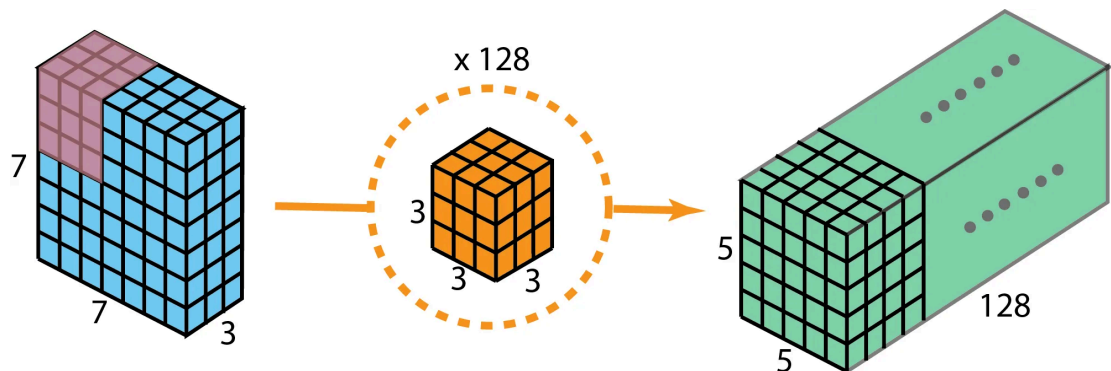


Total numbers of the depthwise separable convolution

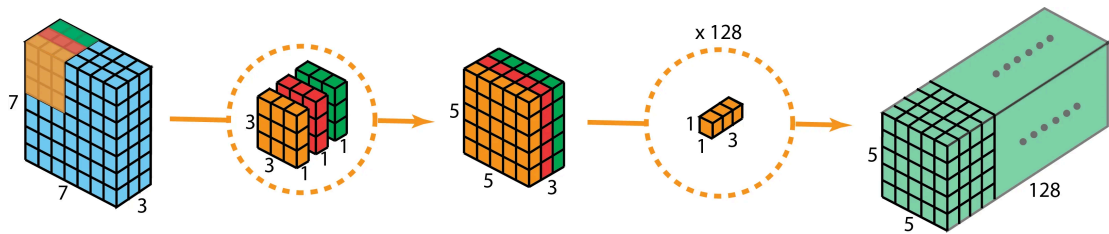
- **Parameters** = $(F \times F \times \text{inC}) + (\text{inC} \times \text{outC})$
- **Operations** = $(F \times F \times \text{inC} \times S \times S) + (\text{inC} \times S \times S \times \text{outC})$
- **Example:** $S = 128, F = 3, \text{inC} = 3, \text{outC} = 16$
 - **Parameters:** $(3 \times 3 \times 3) + (3 \times 16) = 75$
 - **Operations:** $(3 \times 3 \times 3 \times 128 \times 128) + (3 \times 128 \times 128 \times 16) = 1.228.800$

In brief:

- In the normal convolution, we are transforming the image **outC** (128) times.
- **Operations** = $F \times F \times \text{inC} \times [S - 2] \times [S - 2] \times \text{outC}$
 - For the given example: $3 \times 3 \times 3 \times 5 \times 5 \times 128 = 86.400$



- In the separable convolution, we only really transform the image once — in the depthwise convolution.
- Then, we take the transformed image and simply elongate it to **outC** (128) channels. Without having to transform the image over and over again, we can save up on computational power.



- Operations = ($F \times F \times \text{inC} \times [S-2] \times [S-2]$) + ($\text{inC} \times [S-2] \times [S-2] \times \text{outC}$)
 - Example: $(3 \times 3 \times 3 \times 5 \times 5) + (3 \times 5 \times 5 \times 128) = 675 + 9.600 = 10.275$
- **Advantages:**
 - Depthwise separable convolution requires significantly fewer parameters and involves fewer computations compared to regular convolution, while having comparable representational power.
 - It results in smaller models that converge faster and are less prone to overfitting.
 - These advantages become especially important when you're training small models from scratch on limited data.
 - This efficiency make depthwise separable convolution ideal for mobile computer vision models such as Google's MobileNet.
- **Disadvantages:**
 - Because it reduces the number of parameters in a convolution, if your network is already small, you might end up with too few parameters and your network might fail to properly learn during training.

3.11.4. Depthwise separable convolutions in Keras

- In Keras we can apply depthwise separable convolutions using the layer: `keras.layers.SeparableConv2D`
- Let's create **a model with data augmentation, residual connections and depthwise separable convolutions** and apply it to the **cats and dogs example**.

```
In [12]: data_augmentation = keras.Sequential(
[
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.1),
    layers.RandomZoom(0.2),
]
)
```

- After the data augmentation and rescaling layer we put a normal Convolutional 2D with 32 filters.
- Since separable convolutional layers only have one spatial filter per input channel, we should avoid using them after layers that have too few channels, such as the input layer.

```

In [13]: inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)

x = layers.Rescaling(1./255)(x)
x = layers.Conv2D(filters=32, kernel_size=5, use_bias=False)(x)

for size in [32, 64, 128, 256, 512]:
    residual = x

    x = layers.BatchNormalization()(x)
    x = layers.Activation("relu")(x)
    x = layers.SeparableConv2D(size, 3, padding="same", use_bias=False)(x)

    x = layers.BatchNormalization()(x)
    x = layers.Activation("relu")(x)
    x = layers.SeparableConv2D(size, 3, padding="same", use_bias=False)(x)

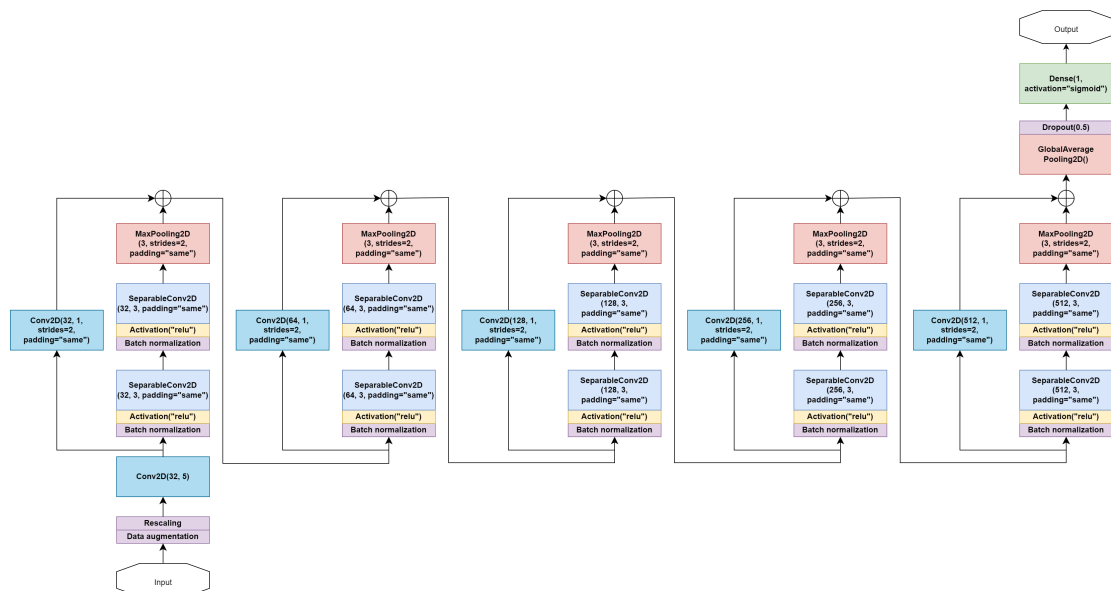
    x = layers.MaxPooling2D(3, strides=2, padding="same")(x)

    residual = layers.Conv2D(size, 1, strides=2, padding="same", use_bias=False)
    x = layers.add([x, residual])

x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs=inputs, outputs=outputs)

```

- We have created a deep CNN.



```

In [14]: model.summary()

```

Model: "functional_3"

Layer (type)	Output Shape	Param #	Connected to
input_layer_3 (InputLayer)	(None , 180, 180, 3)	0	-
sequential_1 (Sequential)	(None , 180, 180, 3)	0	input_layer_3
rescaling (Rescaling)	(None , 180, 180, 3)	0	sequential_1
conv2d_203 (Conv2D)	(None , 176, 176, 32)	2,400	rescaling[0]
batch_normalization_203 (BatchNormalization)	(None , 176, 176, 32)	128	conv2d_203
activation_203 (Activation)	(None , 176, 176, 32)	0	batch_normalization_203
separable_conv2d_203 (SeparableConv2D)	(None , 176, 176, 32)	1,312	activation_203
batch_normalization_204 (BatchNormalization)	(None , 176, 176, 32)	128	separable_conv2d_203
activation_204 (Activation)	(None , 176, 176, 32)	0	batch_normalization_204
separable_conv2d_204 (SeparableConv2D)	(None , 176, 176, 32)	1,312	activation_204
max_pooling2d_4 (MaxPooling2D)	(None , 88, 88, 32)	0	separable_conv2d_204
conv2d_204 (Conv2D)	(None , 88, 88, 32)	1,024	conv2d_203
add (Add)	(None , 88, 88, 32)	0	max_pooling2d_4 conv2d_204
batch_normalization_205 (BatchNormalization)	(None , 88, 88, 32)	128	add[0][0]
activation_205 (Activation)	(None , 88, 88, 32)	0	batch_normalization_205
separable_conv2d_205 (SeparableConv2D)	(None , 88, 88, 64)	2,336	activation_205
batch_normalization_206 (BatchNormalization)	(None , 88, 88, 64)	256	separable_conv2d_205
activation_206 (Activation)	(None , 88, 88, 64)	0	batch_normalization_206
separable_conv2d_206 (SeparableConv2D)	(None , 88, 88, 64)	4,672	activation_206
max_pooling2d_5 (MaxPooling2D)	(None , 44, 44, 64)	0	separable_conv2d_206
conv2d_205 (Conv2D)	(None , 44, 44, 64)	2,048	add[0][0]

	64)		
add_1 (Add)	(None, 44, 44, 64)	0	max_pooling2d_5 (MaxPooling2D)
batch_normalization_206 (BatchNormalization)	(None, 44, 44, 64)	256	add_1[0][0]
activation_207 (Activation)	(None, 44, 44, 64)	0	batch_normalization_206
separable_conv2d_4 (SeparableConv2D)	(None, 44, 44, 128)	8,768	activation_207
batch_normalization_207 (BatchNormalization)	(None, 44, 44, 128)	512	separable_conv2d_4
activation_208 (Activation)	(None, 44, 44, 128)	0	batch_normalization_207
separable_conv2d_5 (SeparableConv2D)	(None, 44, 44, 128)	17,536	activation_208
max_pooling2d_6 (MaxPooling2D)	(None, 22, 22, 128)	0	separable_conv2d_5
conv2d_206 (Conv2D)	(None, 22, 22, 128)	8,192	add_1[0][0]
add_2 (Add)	(None, 22, 22, 128)	0	max_pooling2d_6
batch_normalization_208 (BatchNormalization)	(None, 22, 22, 128)	512	add_2[0][0]
activation_209 (Activation)	(None, 22, 22, 128)	0	batch_normalization_208
separable_conv2d_6 (SeparableConv2D)	(None, 22, 22, 256)	33,920	activation_209
batch_normalization_209 (BatchNormalization)	(None, 22, 22, 256)	1,024	separable_conv2d_6
activation_210 (Activation)	(None, 22, 22, 256)	0	batch_normalization_209
separable_conv2d_7 (SeparableConv2D)	(None, 22, 22, 256)	67,840	activation_210
max_pooling2d_7 (MaxPooling2D)	(None, 11, 11, 256)	0	separable_conv2d_7
conv2d_207 (Conv2D)	(None, 11, 11, 256)	32,768	add_2[0][0]
add_3 (Add)	(None, 11, 11, 256)	0	max_pooling2d_7
batch_normalization_210 (BatchNormalization)	(None, 11, 11, 256)	1,024	add_3[0][0]
activation_211 (Activation)	(None, 11, 11, 256)	0	batch_normalization_210

separable_conv2d_8 (SeparableConv2D)	(None , 11, 11, 512)	133,376	activation_212
batch_normalization_8 (BatchNormalization)	(None , 11, 11, 512)	2,048	separable_conv2d_8
activation_212 (Activation)	(None , 11, 11, 512)	0	batch_normalization_8
separable_conv2d_9 (SeparableConv2D)	(None , 11, 11, 512)	266,752	activation_212
max_pooling2d_8 (MaxPooling2D)	(None , 6, 6, 512)	0	separable_conv2d_9
conv2d_208 (Conv2D)	(None , 6, 6, 512)	131,072	add_3[0][0]
add_4 (Add)	(None , 6, 6, 512)	0	max_pooling2d_8
global_average_pooling2d_8 (GlobalAveragePooling2D)	(None , 512)	0	add_4[0][0]
dropout_1 (Dropout)	(None , 512)	0	global_average_pooling2d_8
dense_2 (Dense)	(None , 1)	513	dropout_1[0]

Total params: 721,857 (2.75 MB)

Trainable params: 718,849 (2.74 MB)

Non-trainable params: 3,008 (11.75 KB)

```
In [15]: # All together
model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])

callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="all_together.keras",
        save_best_only=True,
        monitor="val_loss")
]
history = model.fit(
    train_dataset,
    epochs=100,
    validation_data=validation_dataset,
    callbacks=callbacks)
```

Epoch 1/100
63/63  47s 708ms/step - accuracy: 0.5587 - loss: 0.7857 - val
_accuracy: 0.5000 - val_loss: 0.6930
Epoch 2/100
63/63  45s 710ms/step - accuracy: 0.5885 - loss: 0.6586 - val
_accuracy: 0.4990 - val_loss: 0.7118
Epoch 3/100
63/63  45s 711ms/step - accuracy: 0.6081 - loss: 0.6632 - val
_accuracy: 0.5000 - val_loss: 0.7128
Epoch 4/100
63/63  44s 706ms/step - accuracy: 0.6295 - loss: 0.6430 - val
_accuracy: 0.5000 - val_loss: 0.7096
Epoch 5/100
63/63  45s 708ms/step - accuracy: 0.6872 - loss: 0.6244 - val
_accuracy: 0.5000 - val_loss: 0.7151
Epoch 6/100
63/63  45s 707ms/step - accuracy: 0.6786 - loss: 0.6046 - val
_accuracy: 0.5020 - val_loss: 0.7951
Epoch 7/100
63/63  45s 709ms/step - accuracy: 0.6959 - loss: 0.5911 - val
_accuracy: 0.5230 - val_loss: 0.7979
Epoch 8/100
63/63  45s 708ms/step - accuracy: 0.7059 - loss: 0.5525 - val
_accuracy: 0.5250 - val_loss: 0.8635
Epoch 9/100
63/63  44s 704ms/step - accuracy: 0.7152 - loss: 0.5687 - val
_accuracy: 0.5530 - val_loss: 0.7855
Epoch 10/100
63/63  45s 710ms/step - accuracy: 0.7382 - loss: 0.5434 - val
_accuracy: 0.5930 - val_loss: 0.7958
Epoch 11/100
63/63  45s 709ms/step - accuracy: 0.7422 - loss: 0.5333 - val
_accuracy: 0.6300 - val_loss: 0.7612
Epoch 12/100
63/63  44s 706ms/step - accuracy: 0.7440 - loss: 0.5317 - val
_accuracy: 0.5970 - val_loss: 0.7010
Epoch 13/100
63/63  45s 710ms/step - accuracy: 0.7532 - loss: 0.5251 - val
_accuracy: 0.7270 - val_loss: 0.5356
Epoch 14/100
63/63  45s 710ms/step - accuracy: 0.7839 - loss: 0.4859 - val
_accuracy: 0.7140 - val_loss: 0.6499
Epoch 15/100
63/63  44s 706ms/step - accuracy: 0.7820 - loss: 0.4629 - val
_accuracy: 0.6460 - val_loss: 1.0905
Epoch 16/100
63/63  45s 709ms/step - accuracy: 0.7785 - loss: 0.4945 - val
_accuracy: 0.6070 - val_loss: 0.7946
Epoch 17/100
63/63  45s 709ms/step - accuracy: 0.7883 - loss: 0.4544 - val
_accuracy: 0.6040 - val_loss: 1.0813
Epoch 18/100
63/63  45s 707ms/step - accuracy: 0.8283 - loss: 0.4208 - val
_accuracy: 0.7190 - val_loss: 0.8093
Epoch 19/100
63/63  46s 725ms/step - accuracy: 0.8247 - loss: 0.3993 - val
_accuracy: 0.6690 - val_loss: 0.7708
Epoch 20/100
63/63  45s 709ms/step - accuracy: 0.8034 - loss: 0.4339 - val
_accuracy: 0.7240 - val_loss: 0.7113

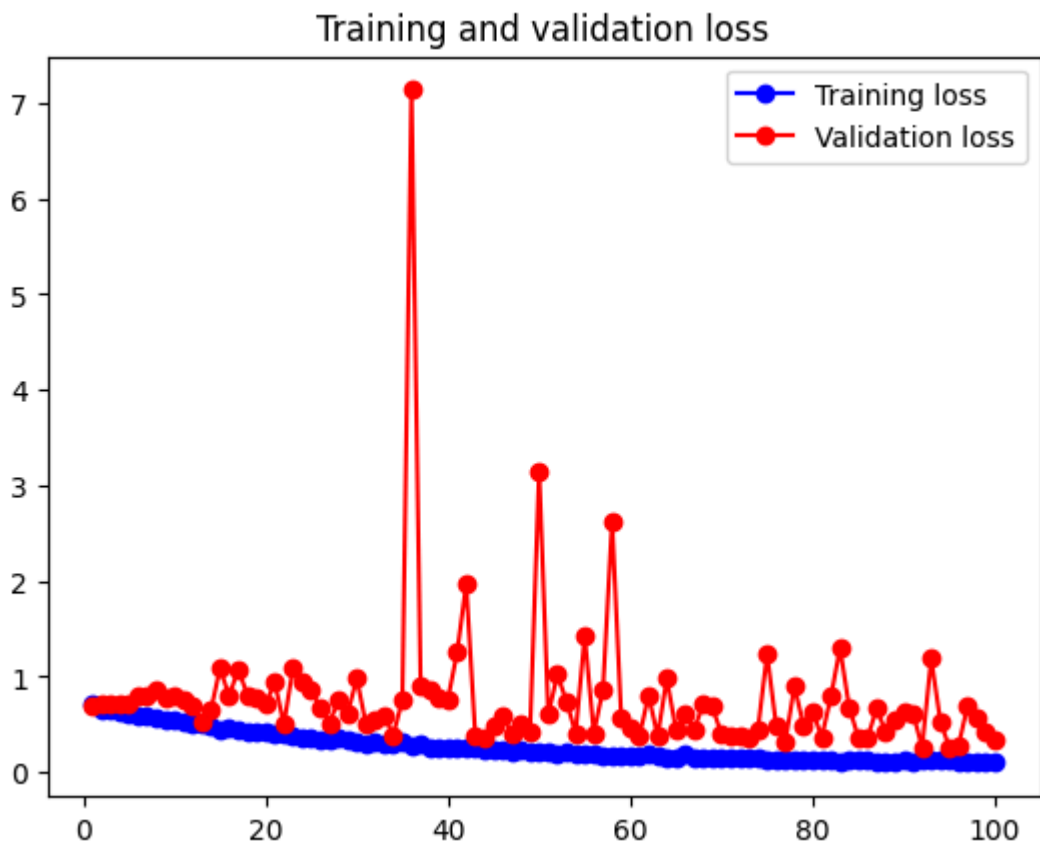
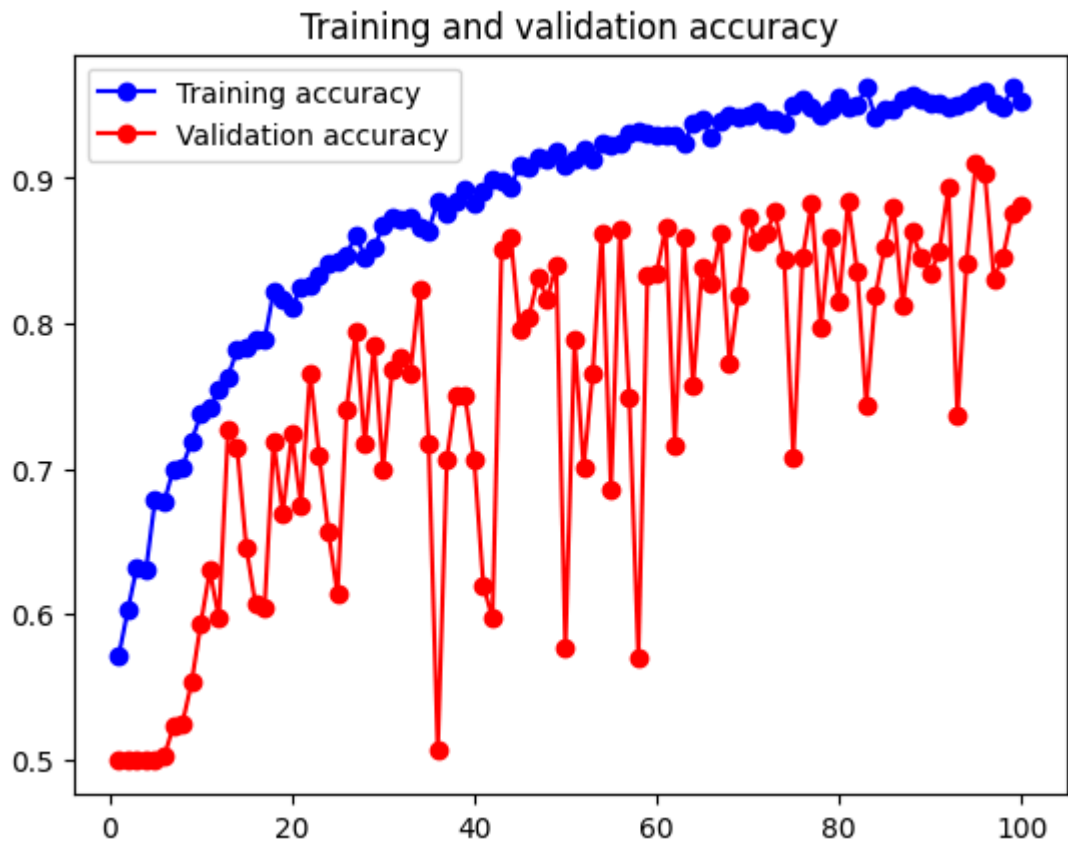
Epoch 21/100
63/63  45s 710ms/step - accuracy: 0.8349 - loss: 0.3902 - val
_accuracy: 0.6750 - val_loss: 0.9427
Epoch 22/100
63/63  45s 713ms/step - accuracy: 0.8319 - loss: 0.4018 - val
_accuracy: 0.7660 - val_loss: 0.5008
Epoch 23/100
63/63  45s 709ms/step - accuracy: 0.8375 - loss: 0.3792 - val
_accuracy: 0.7090 - val_loss: 1.0910
Epoch 24/100
63/63  45s 710ms/step - accuracy: 0.8490 - loss: 0.3589 - val
_accuracy: 0.6570 - val_loss: 0.9549
Epoch 25/100
63/63  45s 707ms/step - accuracy: 0.8480 - loss: 0.3777 - val
_accuracy: 0.6140 - val_loss: 0.8695
Epoch 26/100
63/63  45s 710ms/step - accuracy: 0.8429 - loss: 0.3581 - val
_accuracy: 0.7410 - val_loss: 0.6692
Epoch 27/100
63/63  45s 709ms/step - accuracy: 0.8560 - loss: 0.3463 - val
_accuracy: 0.7940 - val_loss: 0.5153
Epoch 28/100
63/63  44s 704ms/step - accuracy: 0.8313 - loss: 0.3710 - val
_accuracy: 0.7170 - val_loss: 0.7533
Epoch 29/100
63/63  45s 711ms/step - accuracy: 0.8537 - loss: 0.3210 - val
_accuracy: 0.7850 - val_loss: 0.6174
Epoch 30/100
63/63  45s 710ms/step - accuracy: 0.8650 - loss: 0.3194 - val
_accuracy: 0.7000 - val_loss: 0.9822
Epoch 31/100
63/63  44s 706ms/step - accuracy: 0.8795 - loss: 0.2925 - val
_accuracy: 0.7680 - val_loss: 0.5082
Epoch 32/100
63/63  45s 710ms/step - accuracy: 0.8666 - loss: 0.3180 - val
_accuracy: 0.7760 - val_loss: 0.5488
Epoch 33/100
63/63  45s 710ms/step - accuracy: 0.8763 - loss: 0.3123 - val
_accuracy: 0.7660 - val_loss: 0.5994
Epoch 34/100
63/63  45s 707ms/step - accuracy: 0.8448 - loss: 0.3357 - val
_accuracy: 0.8230 - val_loss: 0.3891
Epoch 35/100
63/63  45s 710ms/step - accuracy: 0.8626 - loss: 0.3263 - val
_accuracy: 0.7180 - val_loss: 0.7601
Epoch 36/100
63/63  45s 709ms/step - accuracy: 0.8774 - loss: 0.2929 - val
_accuracy: 0.5060 - val_loss: 7.1333
Epoch 37/100
63/63  45s 708ms/step - accuracy: 0.8716 - loss: 0.3061 - val
_accuracy: 0.7070 - val_loss: 0.9126
Epoch 38/100
63/63  45s 708ms/step - accuracy: 0.8828 - loss: 0.2632 - val
_accuracy: 0.7510 - val_loss: 0.8629
Epoch 39/100
63/63  44s 707ms/step - accuracy: 0.9053 - loss: 0.2513 - val
_accuracy: 0.7500 - val_loss: 0.7741
Epoch 40/100
63/63  45s 709ms/step - accuracy: 0.8774 - loss: 0.2801 - val
_accuracy: 0.7060 - val_loss: 0.7690

Epoch 41/100
63/63  45s 709ms/step - accuracy: 0.8790 - loss: 0.2808 - val
_accuracy: 0.6200 - val_loss: 1.2514
Epoch 42/100
63/63  44s 705ms/step - accuracy: 0.8906 - loss: 0.2593 - val
_accuracy: 0.5970 - val_loss: 1.9673
Epoch 43/100
63/63  45s 709ms/step - accuracy: 0.8954 - loss: 0.2630 - val
_accuracy: 0.8510 - val_loss: 0.3798
Epoch 44/100
63/63  45s 722ms/step - accuracy: 0.8958 - loss: 0.2428 - val
_accuracy: 0.8590 - val_loss: 0.3604
Epoch 45/100
63/63  45s 710ms/step - accuracy: 0.8953 - loss: 0.2704 - val
_accuracy: 0.7960 - val_loss: 0.4898
Epoch 46/100
63/63  45s 710ms/step - accuracy: 0.9160 - loss: 0.2309 - val
_accuracy: 0.8040 - val_loss: 0.5918
Epoch 47/100
63/63  44s 705ms/step - accuracy: 0.9164 - loss: 0.2139 - val
_accuracy: 0.8320 - val_loss: 0.4118
Epoch 48/100
63/63  45s 709ms/step - accuracy: 0.9167 - loss: 0.2178 - val
_accuracy: 0.8170 - val_loss: 0.5146
Epoch 49/100
63/63  45s 708ms/step - accuracy: 0.9204 - loss: 0.2147 - val
_accuracy: 0.8400 - val_loss: 0.4173
Epoch 50/100
63/63  45s 708ms/step - accuracy: 0.9117 - loss: 0.2144 - val
_accuracy: 0.5770 - val_loss: 3.1549
Epoch 51/100
63/63  45s 710ms/step - accuracy: 0.9094 - loss: 0.2141 - val
_accuracy: 0.7890 - val_loss: 0.6036
Epoch 52/100
63/63  45s 707ms/step - accuracy: 0.9216 - loss: 0.1972 - val
_accuracy: 0.7010 - val_loss: 1.0231
Epoch 53/100
63/63  45s 708ms/step - accuracy: 0.9220 - loss: 0.2029 - val
_accuracy: 0.7660 - val_loss: 0.7436
Epoch 54/100
63/63  45s 712ms/step - accuracy: 0.9234 - loss: 0.1949 - val
_accuracy: 0.8620 - val_loss: 0.4104
Epoch 55/100
63/63  44s 706ms/step - accuracy: 0.9101 - loss: 0.2206 - val
_accuracy: 0.6850 - val_loss: 1.4207
Epoch 56/100
63/63  45s 714ms/step - accuracy: 0.9129 - loss: 0.2104 - val
_accuracy: 0.8650 - val_loss: 0.4129
Epoch 57/100
63/63  45s 712ms/step - accuracy: 0.9240 - loss: 0.1961 - val
_accuracy: 0.7490 - val_loss: 0.8650
Epoch 58/100
63/63  45s 707ms/step - accuracy: 0.9336 - loss: 0.1757 - val
_accuracy: 0.5700 - val_loss: 2.6238
Epoch 59/100
63/63  45s 710ms/step - accuracy: 0.9374 - loss: 0.1683 - val
_accuracy: 0.8330 - val_loss: 0.5739
Epoch 60/100
63/63  45s 710ms/step - accuracy: 0.9206 - loss: 0.1992 - val
_accuracy: 0.8350 - val_loss: 0.4700

Epoch 61/100
63/63  44s 705ms/step - accuracy: 0.9324 - loss: 0.1742 - val
_accuracy: 0.8660 - val_loss: 0.3776
Epoch 62/100
63/63  45s 709ms/step - accuracy: 0.9311 - loss: 0.1852 - val
_accuracy: 0.7160 - val_loss: 0.8059
Epoch 63/100
63/63  45s 709ms/step - accuracy: 0.9275 - loss: 0.1638 - val
_accuracy: 0.8590 - val_loss: 0.3794
Epoch 64/100
63/63  45s 710ms/step - accuracy: 0.9424 - loss: 0.1595 - val
_accuracy: 0.7570 - val_loss: 0.9797
Epoch 65/100
63/63  45s 710ms/step - accuracy: 0.9382 - loss: 0.1604 - val
_accuracy: 0.8390 - val_loss: 0.4500
Epoch 66/100
63/63  44s 707ms/step - accuracy: 0.9300 - loss: 0.1831 - val
_accuracy: 0.8270 - val_loss: 0.6153
Epoch 67/100
63/63  45s 711ms/step - accuracy: 0.9413 - loss: 0.1588 - val
_accuracy: 0.8620 - val_loss: 0.4387
Epoch 68/100
63/63  45s 710ms/step - accuracy: 0.9462 - loss: 0.1475 - val
_accuracy: 0.7730 - val_loss: 0.7224
Epoch 69/100
63/63  45s 708ms/step - accuracy: 0.9432 - loss: 0.1439 - val
_accuracy: 0.8200 - val_loss: 0.6865
Epoch 70/100
63/63  45s 710ms/step - accuracy: 0.9386 - loss: 0.1728 - val
_accuracy: 0.8730 - val_loss: 0.3967
Epoch 71/100
63/63  45s 709ms/step - accuracy: 0.9503 - loss: 0.1472 - val
_accuracy: 0.8560 - val_loss: 0.3896
Epoch 72/100
63/63  45s 710ms/step - accuracy: 0.9440 - loss: 0.1414 - val
_accuracy: 0.8620 - val_loss: 0.3808
Epoch 73/100
63/63  45s 712ms/step - accuracy: 0.9479 - loss: 0.1352 - val
_accuracy: 0.8770 - val_loss: 0.3562
Epoch 74/100
63/63  45s 710ms/step - accuracy: 0.9378 - loss: 0.1652 - val
_accuracy: 0.8440 - val_loss: 0.4542
Epoch 75/100
63/63  45s 710ms/step - accuracy: 0.9558 - loss: 0.1208 - val
_accuracy: 0.7080 - val_loss: 1.2358
Epoch 76/100
63/63  45s 709ms/step - accuracy: 0.9508 - loss: 0.1208 - val
_accuracy: 0.8450 - val_loss: 0.4877
Epoch 77/100
63/63  45s 711ms/step - accuracy: 0.9529 - loss: 0.1218 - val
_accuracy: 0.8830 - val_loss: 0.3195
Epoch 78/100
63/63  45s 710ms/step - accuracy: 0.9475 - loss: 0.1315 - val
_accuracy: 0.7970 - val_loss: 0.8993
Epoch 79/100
63/63  45s 712ms/step - accuracy: 0.9426 - loss: 0.1364 - val
_accuracy: 0.8590 - val_loss: 0.4892
Epoch 80/100
63/63  45s 708ms/step - accuracy: 0.9584 - loss: 0.1169 - val
_accuracy: 0.8150 - val_loss: 0.6336

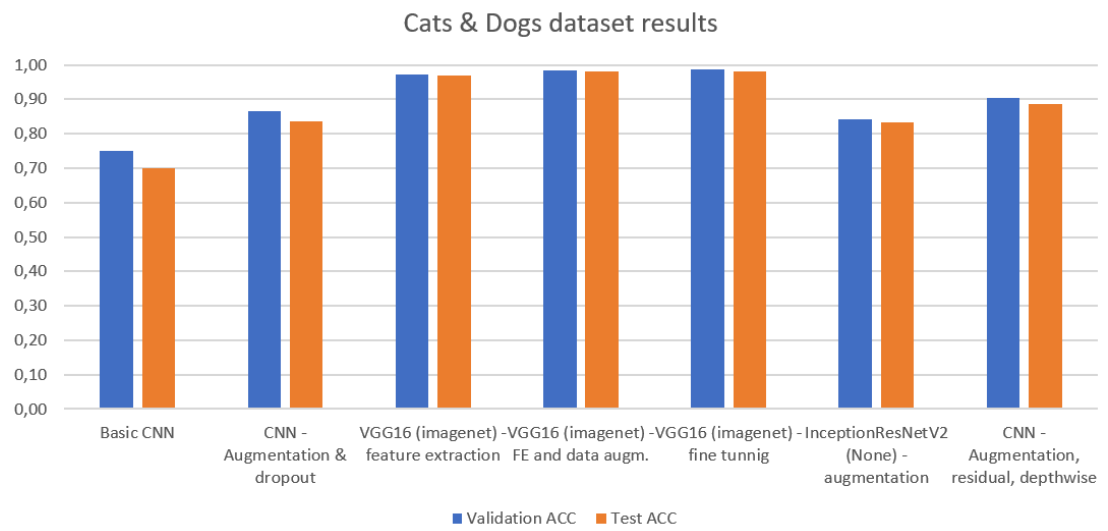
Epoch 81/100
63/63  45s 711ms/step - accuracy: 0.9428 - loss: 0.1527 - val
_accuracy: 0.8840 - val_loss: 0.3700
Epoch 82/100
63/63  45s 707ms/step - accuracy: 0.9443 - loss: 0.1400 - val
_accuracy: 0.8360 - val_loss: 0.7921
Epoch 83/100
63/63  45s 712ms/step - accuracy: 0.9606 - loss: 0.1114 - val
_accuracy: 0.7440 - val_loss: 1.2992
Epoch 84/100
63/63  45s 712ms/step - accuracy: 0.9339 - loss: 0.1434 - val
_accuracy: 0.8190 - val_loss: 0.6828
Epoch 85/100
63/63  44s 707ms/step - accuracy: 0.9427 - loss: 0.1469 - val
_accuracy: 0.8530 - val_loss: 0.3665
Epoch 86/100
63/63  45s 710ms/step - accuracy: 0.9491 - loss: 0.1299 - val
_accuracy: 0.8800 - val_loss: 0.3544
Epoch 87/100
63/63  45s 711ms/step - accuracy: 0.9611 - loss: 0.0979 - val
_accuracy: 0.8120 - val_loss: 0.6656
Epoch 88/100
63/63  45s 708ms/step - accuracy: 0.9558 - loss: 0.1142 - val
_accuracy: 0.8630 - val_loss: 0.4275
Epoch 89/100
63/63  45s 709ms/step - accuracy: 0.9474 - loss: 0.1199 - val
_accuracy: 0.8460 - val_loss: 0.5532
Epoch 90/100
63/63  45s 712ms/step - accuracy: 0.9448 - loss: 0.1419 - val
_accuracy: 0.8340 - val_loss: 0.6273
Epoch 91/100
63/63  45s 717ms/step - accuracy: 0.9479 - loss: 0.1318 - val
_accuracy: 0.8490 - val_loss: 0.6215
Epoch 92/100
63/63  46s 726ms/step - accuracy: 0.9464 - loss: 0.1416 - val
_accuracy: 0.8940 - val_loss: 0.2630
Epoch 93/100
63/63  46s 723ms/step - accuracy: 0.9594 - loss: 0.1021 - val
_accuracy: 0.7360 - val_loss: 1.1888
Epoch 94/100
63/63  45s 720ms/step - accuracy: 0.9484 - loss: 0.1402 - val
_accuracy: 0.8420 - val_loss: 0.5388
Epoch 95/100
63/63  45s 722ms/step - accuracy: 0.9520 - loss: 0.1409 - val
_accuracy: 0.9100 - val_loss: 0.2531
Epoch 96/100
63/63  45s 712ms/step - accuracy: 0.9611 - loss: 0.1113 - val
_accuracy: 0.9030 - val_loss: 0.2837
Epoch 97/100
63/63  45s 711ms/step - accuracy: 0.9491 - loss: 0.1118 - val
_accuracy: 0.8310 - val_loss: 0.6992
Epoch 98/100
63/63  45s 711ms/step - accuracy: 0.9580 - loss: 0.0966 - val
_accuracy: 0.8460 - val_loss: 0.5643
Epoch 99/100
63/63  45s 715ms/step - accuracy: 0.9648 - loss: 0.0852 - val
_accuracy: 0.8760 - val_loss: 0.4321
Epoch 100/100
63/63  45s 711ms/step - accuracy: 0.9543 - loss: 0.1088 - val
_accuracy: 0.8810 - val_loss: 0.3327

```
In [16]: plot(history)
```



```
In [17]: test_model = keras.models.load_model(  
         "all_together.keras")  
test_loss, test_acc = test_model.evaluate(test_dataset)  
print(f"Test accuracy: {test_acc:.3f}")
```

63/63 ————— 9s 141ms/step - accuracy: 0.8982 - loss: 0.2672
Test accuracy: 0.896



In []: